# Supporting Information
# Semiempirical Orbital-Optimized DFT in MOUSE: Spin-Pure Excited-State Surfaces with Analytic Gradients and Hessians

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Working manuscript, 1 November 2026

## 1 Data scope and fixed protocol

This document preserves separately identified model generations. The early tables describe the relative-active-pair qualification; the final screened symmetry-augmented OO-CID assessment is reported in its own section and table below. Numerical evidence from an earlier model is not attributed to the final candidate. The two parent weights are 0.5, the exchange-response scaling is 0.9, and the additive excitation-energy offset is zero. The full and qualified-spectral parents share their quadratic charge/spin correction and differ in their one-body operator. The mixed-spin and triplet energies within each parent use common spatial orbitals.

The 15 regression states were exposed during development. The four final prospective molecules had previous native/shell-model predictions, but the present full/spectral composite protocol was frozen before their new predictions. Reference transition character, rather than TBE proximity, determines assignments. The reported states constitute a curated qualified cohort; the table is not a survey of every attempted state in preceding development campaigns. In particular, its size must not be used as a database-wide success-rate denominator.

The spectral matching uses occupation-sector assignments with absolute overlap at least 0.7 and a best-versus-alternative margin of at least 0.05. Unqualified orbital levels retain native spacings. There is no fitted orbital-extent correction and no multiplication of the orbital gaps by 0.9. The effective exact-exchange response coefficient is 0.225, and the semilocal response is unchanged. The mean occupied energy sets a gauge for spectral shifts, not a fitted shift of excitation energies.

The two relative active-pair targets are squared cross-parent overlaps at the anchor. They remain fixed while the absolute orbital orientations and spectators relax. Targets within $10^{-10}$ of unity use an exact shared-orbital elimination. The final stationarity certificate concerns the coupled Lagrangian; independent parent orbital gradients are not assumed to vanish away from the anchor. Projected absolute orbital curvatures must exceed $10^{-7}\ E_\mathrm{h}$ for certification. The full-CSF and active-character following thresholds are 0.7.

## 2 Complete vertical-energy results

Table S1: All reported vertical excitation energies and signed errors in eV. The final four rows are prospective tests; the preceding rows are exposed regression examples.

| Molecule | State | TBE | OO-DFT | Error |
|---|---|---|---|---|
| Acetaldehyde | $n \to \pi^*$ | 4.304 | 4.277 | -0.027 |
| Acetone | $n \to \pi^*$ | 4.468 | 4.338 | -0.130 |
| Acrolein | $n \to \pi^*$ | 3.723 | 3.771 | +0.048 |
| Cyclopentadienone | $n \to \pi^*$ | 2.951 | 2.953 | +0.002 |
| Formamide | $n \to \pi^*$ | 5.644 | 5.648 | +0.004 |
| Glyoxal | $n \to \pi^*$ | 2.877 | 2.752 | -0.125 |
| Maleimide | $n \to \pi^*$ | 3.810 | 3.643 | -0.167 |
| Nitrosomethane | $n \to \pi^*$ | 1.952 | 1.721 | -0.231 |
| Nitroxyl | $n \to \pi^*$ | 1.743 | 1.485 | -0.258 |
| Pyranone | $n \to \pi^*$ | 4.016 | 4.076 | +0.060 |
| Pyrazine | $n \to \pi^*$ | 4.144 | 3.932 | -0.212 |
| Pyrimidine | $n \to \pi^*$ | 4.451 | 4.218 | -0.233 |
| Tetrazine | $n \to \pi^*$ | 2.462 | 2.060 | -0.402 |
| Uracil | $n \to \pi^*$ | 4.935 | 4.885 | -0.050 |
| Uracil | $\pi \to \pi^*$ | 5.381 | 5.686 | +0.305 |
| Cyanoformaldehyde | $n \to \pi^*$ | 3.817 | 3.989 | +0.172 |
| Cyclopropenone | $n \to \pi^*$ | 4.274 | 4.482 | +0.208 |
| Imidazole | $n \to \pi^*$ | 6.715 | 6.180 | -0.535 |
| Pyridazine | $n \to \pi^*$ | 3.830 | 3.530 | -0.300 |

## 3 Direct dedicated sTDA-xTB comparison

The published dedicated Hamiltonian and parameters are retained. Ground-state xtb4stda wavefunctions and std2 singlet vectors are evaluated at the original XYZs, without geometry optimization or TBE-based root assignment. Cross-basis transition overlaps are normalized only after projection; the norm, overlap, and assignment margin are reported separately in `data/direct_stda_comparison.csv`. All 19 states pass overlap $\geq 0.7$, projected norm $\geq 0.7$, and margin $\geq 0.05$. The small difference between coordinate conversions stored by the legacy executable and the input XYZ is bounded by $2 \times 10^{-6}$ bohr; no geometry is optimized or refitted. Earlier parameter-path setup failures are preserved separately from the successful electronic attempts.

Table S2: Every direct sTDA-xTB comparison. Energies and signed errors are in eV; $q$ is the absolute normalized cross-AO transition overlap.

| Molecule | State | TBE | MOUSE | sTDA-xTB | $\Delta_{\mathrm{sTDA}}$ | $q$ |
|---|---|---|---|---|---|---|
| Acetaldehyde | $n\pi^*$ | 4.304 | 4.277 | 4.454 | +0.150 | 0.915 |
| Acetone | $n\pi^*$ | 4.468 | 4.338 | 4.531 | +0.063 | 0.930 |
| Acrolein | $n\pi^*$ | 3.723 | 3.771 | 3.826 | +0.103 | 0.918 |
| Cyanoformaldehyde | $n\pi^*$ | 3.817 | 3.989 | 3.911 | +0.094 | 0.900 |
| Cyclopentadienone | $n\pi^*$ | 2.951 | 2.953 | 2.840 | -0.111 | 0.923 |
| Cyclopropenone | $n\pi^*$ | 4.274 | 4.482 | 3.838 | -0.436 | 0.902 |
| Formamide | $n\pi^*$ | 5.644 | 5.648 | 6.039 | +0.395 | 0.853 |
| Glyoxal | $n\pi^*$ | 2.877 | 2.752 | 2.007 | -0.870 | 0.924 |

| Molecule | State | TBE | MOUSE | sTDA-xTB | $\Delta_{\mathrm{sTDA}}$ | $q$ |
|---|---|---|---|---|---|---|
| Imidazole | $n\pi^*$ | 6.715 | 6.180 | 7.277 | +0.562 | 0.739 |
| Maleimide | $n\pi^*$ | 3.810 | 3.643 | 3.654 | -0.156 | 0.931 |
| Nitrosomethane | $n\pi^*$ | 1.952 | 1.721 | 0.492 | -1.460 | 0.950 |
| Nitroxyl | $n\pi^*$ | 1.743 | 1.485 | 0.237 | -1.506 | 0.955 |
| Pyranone | $n\pi^*$ | 4.016 | 4.076 | 4.710 | +0.694 | 0.913 |
| Pyrazine | $n\pi^*$ | 4.144 | 3.932 | 4.213 | +0.069 | 0.910 |
| Pyridazine | $n\pi^*$ | 3.830 | 3.530 | 3.810 | -0.020 | 0.921 |
| Pyrimidine | $n\pi^*$ | 4.451 | 4.218 | 4.684 | +0.233 | 0.932 |
| Tetrazine | $n\pi^*$ | 2.462 | 2.060 | 1.753 | -0.709 | 0.946 |
| Uracil | $n\pi^*$ | 4.935 | 4.885 | 5.760 | +0.825 | 0.892 |
| Uracil | $\pi\pi^*$ | 5.381 | 5.686 | 5.210 | -0.171 | 0.884 |

# 4 Physical structural accuracy and weight precheck

The following internal-coordinate references are from the published CCSDR(3)/def2-TZVPP supporting tables (DOI 10.1021/acs.jctc.7b00921): acetaldehyde S19, acetone S20, and maleimide S27. The geometry definitions and reference provenance are retained in `data/geometry_reference_comparison.json`. No structural reference was used to fit the energy model.

Table S3: Internal coordinates of MOUSE excited-state minima against CCSDR(3)/def2-TZVPP. These comparisons assess physical accuracy separately from derivative correctness.

| Molecule | Coordinate | MOUSE | Reference | Error | Unit |
|---|---|---|---|---|---|
| Acetaldehyde | C-O | 1.312 | 1.331 | -0.019 | Å |
| Acetaldehyde | C-C | 1.488 | 1.508 | -0.020 | Å |
| Acetaldehyde | C-Hald | 1.098 | 1.091 | +0.007 | Å |
| Acetaldehyde | C-C-O | 116.841 | 115.200 | +1.641 | ° |
| Acetaldehyde | C-C-Hald | 113.767 | 119.300 | -5.533 | ° |
| Acetaldehyde | pyramidalization | 41.651 | 38.700 | +2.951 | ° |
| Acetone | C-O | 1.319 | 1.336 | -0.017 | Å |
| Acetone | C-C | 1.489 | 1.507 | -0.018 | Å |
| Acetone | C-C-C | 117.761 | 119.500 | -1.739 | ° |
| Acetone | pyramidalization | 30.555 | 42.300 | -11.745 | ° |
| Maleimide | N-C | 1.378 | 1.404 | -0.026 | Å |
| Maleimide | C-O | 1.244 | 1.248 | -0.004 | Å |
| Maleimide | C-C | 1.426 | 1.432 | -0.006 | Å |
| Maleimide | C=C | 1.371 | 1.386 | -0.015 | Å |
| Maleimide | N-H | 1.012 | 1.006 | +0.006 | Å |
| Maleimide | C-H | 1.072 | 1.074 | -0.002 | Å |
| Maleimide | C-N-C | 111.090 | 109.800 | +1.290 | ° |

The weight table uses independently stationary vertical parents only. It cannot substitute for reoptimizing a weighted off-anchor coupled problem. The production weight remains 0.5. The separate harmonic nitroxyl precheck is retained in `data/hno_harmonic_comparison.json`; its frequency-order correspondence remains provisional until mode-vector confirmation.

Table S4: Algebraic weight sensitivity for the 19 independently stationary vertical-anchor parent gaps. This is not a test of the weighted off-anchor potential or its Hessian. The production weights remain 0.5/0.5.

| $w_F$ | MAE (eV) | RMSE (eV) | Max. error (eV) |
|---|---|---|---|
| 0.00 | 0.210 | 0.257 | 0.645 |
| 0.25 | 0.179 | 0.211 | 0.590 |
| 0.50 | 0.183 | 0.229 | 0.535 |
| 0.75 | 0.239 | 0.300 | 0.583 |
| 1.00 | 0.308 | 0.396 | 0.958 |

# 5 Reference data provenance

The machine-readable file `data/tbe_provenance.json` associates each state with its archived QUEST TBE method. The target basis is aug-cc-pVTZ (AVTZ); AVDZ denotes aug-cc-pVDZ. These are theoretical vertical references, not measured absorption maxima. The full input metadata and inherited state assignments are in `reference_provenance.json`. The DFT reference basis is aug-cc-pVDZ; it is distinct from the target basis of the TBE. These values are used for reporting errors, not constructing the semiempirical energy.
**Acetaldehyde (n_pi):** CCSDTQ/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
**Acetone (n_pi):** CCSDTQ/6-31+G(d) + [CC4/AVDZ - CC4/6-31+G(d)] + [CCSDT/AVTZ - CCSDT/AVDZ].
**Acrolein (n_pi):** CCSDTQ/6-31+G(d) + [CC4/AVDZ - CC4/6-31+G(d)] + [CCSDT/AVTZ - CCSDT/AVDZ].
**Cyclopentadienone (n_pi):** CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
**Formamide (n_pi):** CCSDTQ/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
**Glyoxal (n_pi):** CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
**Maleimide (n_pi):** CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/AVDZ].
**Nitrosomethane (n_pi):** CCSDTQ/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
**Nitroxyl (n_pi):** CCSDTQP/6-31+G(d) + [CCSDTQ/AVTZ - CCSDTQ/6-31+G(d)].
**Pyranone (n_pi):** CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
**Pyrazine (n_pi):** CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
**Pyrimidine (n_pi):** CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
**Tetrazine (n_pi):** CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
**Uracil (n_pi):** CCSDT/6-31+G(d) + [CC3/AVTZ - CC3/6-31+G(d)].
**Uracil (pi_pi):** CCSDT/6-31+G(d) + [CC3/AVTZ - CC3/6-31+G(d)].
**Cyanoformaldehyde (n_pi):** CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
**Cyclopropenone (n_pi):** CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
**Imidazole (n_pi):** CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
**Pyridazine (n_pi):** CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].

# 6 Derivative checks and stationarity criteria

The five nuclear minima satisfy a maximum Cartesian gradient below $3 \times 10^{-5}$ $E_\mathrm{h}$/bohr and have positive projected vibrational curvatures. Positive excitation gaps and tracked state character are required as separate checks. Directional checks use central differences of the scalar energy and analytic gradient at $\pm hv$, with $h = 2 \times 10^{-4}$ and $10^{-4}$ bohr. The reported Hessian-vector discrepancy is the largest component of the difference from the analytic Hessian acting on the same direction. No finite differences enter the production gradient or Hessian.

At $h = 10^{-4}$ bohr, complete Cartesian-column checks yield:

| Molecule | Columns | Maximum Hessian error | Scalar-gradient error |
|---|---|---|---|
| Nitroxyl | 9 | $1.1512 \times 10^{-8}$ | $3.6434 \times 10^{-9}$ |
| Formamide | 18 | $4.6669 \times 10^{-9}$ | $2.2575 \times 10^{-9}$ |

Hessian errors are in $E_\mathrm{h}/\mathrm{bohr}^2$; scalar-gradient errors are in $E_\mathrm{h}/\mathrm{bohr}$. The maximum asymmetry of the numerical Hessians is $8.74 \times 10^{-9}$ and $5.30 \times 10^{-9}$ $E_\mathrm{h}/\mathrm{bohr}^2$, respectively. Full displacement checks and the two-step directional checks for every minimum are recorded in `data/SUMMARY.json`. Translational and rotational modes are removed before vibrational stability is assessed. No experimental-frequency accuracy is inferred from these implementation tests.

# 7 Optimized geometries and projected harmonic frequencies

All coordinates below are in Angstrom, in the original atom order. They describe minima of the constrained composite semiempirical surface. The XYZ files are also supplied in `coordinates/`. These coordinates are not high-level reference geometries.

## 7.1 Acetaldehyde

```
7
acetaldehyde; constrained OO-DFT minimum; coordinates in Angstrom
C   0.2223035093  0.1013277113  0.2334913667
C  -0.9453005961 -0.0251942583 -0.6804940117
O   0.4079243109 -0.8702860896  1.0954507793
H   0.3010447606  1.0841255171  0.7176125575
H  -1.8688390394 -0.3135713164 -0.1481562051
H  -0.7506835894 -0.8350717205 -1.3936582758
H  -1.1076978659  0.9035664200 -1.2182916609
```

Projected harmonic frequencies ($cm^{-1}$): 186.89, 432.23, 667.85, 973.26, 990.28, 1095.13, 1240.79, 1375.25, 1389.50, 1416.35, 1455.67, 2766.35, 2838.16, 2919.28, 3059.54.

## 7.2 Acetone

```
10
acetone; constrained OO-DFT minimum; coordinates in Angstrom
C  -0.5689937111 -0.0533836291 -0.2955897103
C   0.0475047310  1.2845796508 -0.5101289622
C   0.2806363777 -1.2536119814 -0.5275519803
O  -1.8450965398 -0.1685475055 -0.6076164069
H   1.0922037630  1.2729010514 -0.2125706389
H   1.3076035812 -1.0537316303 -0.2352636408
H  -0.0227253198  1.6074367061 -1.5645009174
H  -0.4976868798  2.0329363229  0.0742268604
H   0.2629093176 -1.5736462119 -1.5849234343
H  -0.1140660180 -2.0949327727  0.0514752619
```

Projected harmonic frequencies ($cm^{-1}$): 148.28, 149.63, 270.71, 368.65, 500.42, 852.57, 968.79, 988.72, 1000.93, 1047.16, 1229.45, 1371.00, 1387.39, 1394.90, 1416.89, 1425.45, 1460.92, 1468.31, 2741.86, 2761.53, 2932.39, 2934.40, 3043.80, 3048.12.

## 7.3 Formamide

```
6
formamide; constrained OO-DFT minimum; coordinates in Angstrom
C  -0.1991028844 -0.2725711755  0.5163934339
O   1.0835179824 -0.3792440676  0.1254002010
N  -1.0286807403  0.2161588604 -0.4725654836
H  -0.1480202140  0.3692094734  1.4479826708
H  -0.5794619260  0.4820208966 -1.3402912231
H  -1.8635741275 -0.3394673884 -0.6301074391
```

Projected harmonic frequencies ($cm^{-1}$): 324.38, 489.56, 567.75, 902.41, 1043.32, 1172.31, 1242.56, 1256.85, 1491.02, 2377.42, 3308.98, 3430.06.

### 7.4 Maleimide

```
10
maleimide; constrained OO-DFT minimum; coordinates in Angstrom
C   0.0000000000  1.1364919551  0.0927366553
C   0.0000000000 -1.1364919551  0.0927366554
C  -0.0000000000  0.6854283480 -1.2605427879
C   0.0000000000 -0.6854283480 -1.2605427880
N  -0.0000000000  0.0000000000  0.8725164373
O   0.0000000000  2.2842746107  0.5713837699
O   0.0000000000 -2.2842746107  0.5713837699
H   0.0000000000  0.0000000000  1.8848555803
H  -0.0000000000  1.3571912082 -2.0958657911
H  -0.0000000000 -1.3571912082 -2.0958657910
```

Projected harmonic frequencies ($cm^{-1}$): 179.17, 288.09, 385.45, 502.10, 527.27, 554.29, 601.09, 618.76, 648.38, 676.37, 705.14, 774.17, 992.59, 1029.27, 1049.28, 1165.93, 1220.57, 1252.23, 1312.02, 1486.37, 1636.37, 3169.72, 3180.16, 3405.90.

### 7.5 Nitroxyl

```
3
nitroxyl; constrained OO-DFT minimum; coordinates in Angstrom
O   0.0826117682  0.0000000000  1.1867248604
N  -0.1842703289 -0.0000000000 -0.0200741732
H   0.6016583607 -0.0000000000 -0.6666508772
```

Projected harmonic frequencies ($cm^{-1}$): 1261.41, 1592.58, 3301.05.

## 8 Timing interpretation

Earlier values labelled TDA were built with a symmetrized static response and omitted the nonlocal exchange difference required for the TDA $A$ matrix. Their timing ratios and TDA accuracy claims are withdrawn. The archival measurements are explicitly identified in `data/archival_static_response_timing.json`; the corresponding CSV fields are labelled archival static response. Corrected TDA uses the corrected GFN2 ground orbitals, with effective exact exchange 0.225, and agrees with the independent PySCF matrix within $3.6 \times 10^{-15}$ $E_h$.

The batched analytic Jacobian comparison uses the identical retained 44-AO nitropyridine checkpoint and two Oracle threads: 4.13837 versus 0.241612 s, maximum elementwise difference $8.88 \times 10^{-16}$. This 17.13-fold kernel speedup excludes reference preparation and does not establish an end-to-end advantage. The latest response/ROKS/relative-pair/trial-guard regression suite passes 31 tests on Oracle, including trusted Newton acceleration for resolved soft saddles.

The reduced extension initializes stationary parents from independently assigned NTO and native seeds. It applies local CSF overlap 0.90, attainable single-pair transition fidelity squared above one half, occupied/virtual norms 0.7 and an available-root projected margin of 0.05. The ratio to the leading singular value is a representability diagnostic, not a state probability. Absolute active-orbital anchor overlap is reported diagnostically when a fixed reference transition is supplied. Stationarity and curvature criteria remain unchanged. This assignment amendment is separated from the original 19-state data and from the subsequently locked accuracy assessment. The reduced scope is 14 states in ten molecules, selected after baseline convergence outcomes; no excluded state is silently removed from the original ledger.

## 9 Completed reduced extension

All 14 predictions and matched TDA assignments were locked before label opening. The scope was reduced after inspecting convergence and is not wholly blind. The original prediction lock, scoring output, label-opening record and full scope ledger accompany the document in `data/reduced_extension/`. Failed or unresolved earlier attempts remain in the calculation archive.

Table S5: Every reduced-extension energy (eV). The original prediction lock precedes label opening. TDA uses the corrected GFN2 ground orbitals.

| ID / molecule | Family | TBE | OO | TDA-F | TDA-Q | TDA mean |
|---|---|---|---|---|---|---|
| B001 Diazirine | $n\pi^*$ | 4.093 | 4.242 | 5.247 | 4.230 | 4.739 |
| B004 Nitropyridine_N-oxide | $n\pi^*$ | 4.395 | 4.263 | 4.096 | 4.515 | 4.305 |
| B005 Phthalimide | $n\pi^*$ | 4.186 | 4.070 | 4.029 | 4.004 | 4.016 |
| B007 Adenine | $n\pi^*$ | 5.304 | 4.805 | 4.245 | 5.192 | 4.719 |
| B011 Aza-naphthalene | $n\pi^*$ | 5.346 | 5.279 | 5.054 | 5.507 | 5.281 |
| B014 Adenine | $\pi\pi^*$ | 5.191 | 5.449 | 6.192 | 5.494 | 5.843 |
| B015 Bimane_Syn | $\pi\pi^*$ | 3.964 | 3.911 | 5.189 | 4.947 | 5.068 |
| B016 Cyclopentadiene | $\pi\pi^*$ | 6.451 | 8.197 | 8.282 | 6.624 | 7.453 |
| B022 Furan | $\pi\pi^*$ | 6.530 | 7.974 | 8.448 | 6.829 | 7.638 |
| B024 Phthalimide | $\pi\pi^*$ | 4.600 | 5.869 | 5.670 | 4.927 | 5.298 |
| B025 Quinoxaline | CT | 6.225 | 6.250 | 5.718 | 5.933 | 5.825 |
| B029 Nitropyridine_N-oxide | CT | 4.078 | 4.153 | 4.765 | 4.433 | 4.599 |
| B030 Quinoxaline | CT | 4.647 | 5.108 | 6.123 | 4.562 | 5.343 |
| B039 Benzoquinone | Rydberg | 7.083 | 7.681 | 8.582 | 7.212 | 7.897 |

## 10 Complete furan multi-CSF Hessian check

This check concerns the experimental full-relative-active-frame multi-CSF functional, separately from the original single-CSF minima and derivative checks. All 27 Cartesian columns are included; the maximum error is $2.2852 \times 10^{-8}$ $E_\mathrm{h}/\mathrm{bohr}^2$. Electronic solutions are reoptimized at each displaced point, and all three ground-reference terms are included. Completed points and columns are retained for restart.

Table S6: Complete 27-column furan total excited-state Hessian validation. Analytic constrained orbital/CI response and all three ground-reference derivatives are included. Central differences reoptimize the state at each displaced geometry with step $2 \times 10^{-4}$ bohr. Units are $E_h$/bohr for the gradient check and $E_h/\mathrm{bohr}^2$ for the Hessian.

| Cartesian column (zero based) | Energy–gradient error | Max. Hessian error |
|---|---|---|
| 0 | 1.35e-15 | 1.25e-08 |
| 1 | 1.63e-09 | 1.65e-08 |
| 2 | 8.17e-09 | 1.52e-08 |
| 3 | 7.11e-11 | 1.25e-08 |
| 4 | 2.07e-09 | 1.65e-08 |
| 5 | 8.33e-09 | 1.52e-08 |

| Cartesian column (zero based) | Energy–gradient error | Max. Hessian error |
|---|---|---|
| 6 | 1.87e-10 | 9.93e-09 |
| 7 | 7.82e-09 | 1.83e-08 |
| 8 | 6.42e-09 | 2.28e-08 |
| 9 | 1.11e-10 | 9.93e-09 |
| 10 | 7.84e-09 | 1.83e-08 |
| 11 | 6.68e-09 | 2.29e-08 |
| 12 | 1.38e-10 | 1.89e-09 |
| 13 | 2.22e-11 | 8.80e-09 |
| 14 | 3.88e-09 | 2.80e-09 |
| 15 | 2.22e-10 | 2.13e-09 |
| 16 | 5.37e-09 | 1.16e-08 |
| 17 | 2.19e-10 | 2.98e-09 |
| 18 | 7.99e-11 | 2.13e-09 |
| 19 | 5.20e-09 | 1.16e-08 |
| 20 | 1.57e-10 | 2.98e-09 |
| 21 | 6.66e-11 | 1.56e-09 |
| 22 | 9.31e-10 | 2.77e-09 |
| 23 | 3.33e-09 | 9.41e-09 |
| 24 | 2.93e-10 | 1.56e-09 |
| 25 | 7.62e-10 | 2.78e-09 |
| 26 | 3.14e-09 | 9.41e-09 |

## 11 Reproducibility and automatic integration

The final remote qualification used `oracle.sns.it`. A transfer audit checked 561 result files and 30 source files; the remote test suite reports 86 passing tests. Cold and resumed executions of the integrated example differ by $1.7764 \times 10^{-15}$ $E_\mathrm{h}$. The copied `data/PROTOCOL.json` identifies the validated source files by SHA256. The source hashes refer to the calculation campaign; the manuscript-generation manifest separately identifies the document inputs and outputs.

The explicit interface selects the following settings:

```
native_excited_method = "roks_50_50"
native_roks_composite_orbital_coupling = "relative_active_pair"
native_roks_composite_following = "character_newton_index"
native_roks_composite_reference_policy = "geometry_only"
native_roks_composite_ground_surface = "native_plus_gaps"
native_roks_composite_component_tracking = "active_pair"
```

The example source is `examples/roks_relative_pair.py` in the MOUSE repository. Response paths, two certified seed paths, promotions, and parent saddle indices are supplied explicitly. The accepted generation has precedence over cold seeds. It stores both CSFs, relative targets, multipliers, three restricted ground references, model fingerprints, and the anchor. Defaults for other methods are not changed.

The manuscript-generation script `build_assets.py` performs lightweight data aggregation and plotting only. It can be run from the associated MOUSE working tree to regenerate all tables, the validation figure, optimized XYZ files, and TBE method provenance. The accompanying CSV provides all 19 OO-DFT values and explicitly labelled archival static-response controls. The completed reduced-extension corrected TDA and OO values are reported separately below. The four-molecule comparison in the main text is the designated prospective comparison; additional exposed TDA values in the CSV must retain that exposure label.

## 12 Public multi-CSF interface qualification

The explicit research interface is

```
native_excited_method = "multicsf_50_50"
native_multicsf_model_path = Path("../model.json")
```

The model defines promotions, rank, reference transitions and the full relative frame. The example is `examples/multicsf_oo_dft.py`. Its input hashes are checked before optimizer-cache access. Accepted generations store both wavefunctions, CI coefficients, three ground references and derivatives; trial evaluations do not publish a new generation. A relocated complete model restores the same accepted state. Checkpoint corruption and modified model arrays are rejected. These checks and 46 distinct regression tests ran on Oracle with two threads. No electronic job ran on the laptop.

The cold full-Hessian interface call takes 98.99 s; restoring the accepted state and its already stored Hessian takes 0.613 s. The latter is not a fresh-Hessian timing. Surface evaluation is instrumented against SCC calls and invokes no DFT/TDA solver. Preparation still requires the archived DFT response and reference transition. All source/result hashes, native-extension hash, logs and portable furan inputs are recorded in the multi-CSF integration qualification archive. The original single-CSF automatic interface above remains a separate model.

The response-only $\lambda$ diagnostic uses 0.85, 0.90 and 0.95, with the one-body DFT reference, semilocal contribution and bare off-diagonal CSF couplings held fixed. Vertical weight and fixed-reference rank perturbations are separate algebraic diagnostics; no new molecular minimum or weighted Hessian is inferred. The actual displaced-reference rank diagnostic is reported below; full F/Q energy recalibration and independent validation remain open.

## 13 Symmetry-assigned TD-DFTB and ΔDFTB comparison

All electronic calculations in this table were already completed on Oracle; the new audit only reads stored wavefunctions, transition weights, occupations and energies. DFTB2 uses mio-1-1 parameters. The auxiliary Gaussian $s/p$ shells define the same angular group representation and are never used to compute an electronic energy or a cross-model radial overlap. Reference axes are reused explicitly to avoid irrep-name changes from equivalent axis conventions. A TD root requires more than 0.999 of the reported transition weight in its assigned irrep. The reference irrep is obtained from its AO transition density, allowing orbital degeneracy and mixing. The ΔDFTB final occupied spin orbitals must have irrep purity above 0.999; their product gives the broken-symmetry determinant's spatial irrep. Only globally first reference singlets are considered for this lowest-state method.

The 19-state TD-DFTB MAE is 0.54547 eV. For the 15 eligible completed ΔDFTB states, its MAE is 0.31353 eV and the same-subset MOUSE MAE is 0.17174 eV. On the four previous prospective cases, TD-DFTB gives 1.05835 eV MAE versus 0.30363 eV for MOUSE; on the three eligible prospective ΔDFTB cases, the respective MAEs are 0.23380 and 0.22655 eV. The original calculations, including unsuccessful SCC attempts for uracil and cyclopentadienone, are retained. No failed calculation is replaced by an assigned energy. Additional bounded Anderson and linear-mixing recoveries retain the same mio-1-1 Hamiltonian, integer occupations and $10^{-10}$ SCC tolerance; both incomplete cases remain unresolved. The backend rejects binary charge restart for this method, so the final recoveries supply the previously converged atomic ground charges through the supported InitialCharges input. All 112 recovery artifacts, including input rejections and unsuccessful SCC histories, are retained separately from the 254 original audit inputs/results. Neither the group-theoretical audit nor the arithmetic score is a cross-model transition-overlap qualification or a new independent validation set.

Table S7: TD-DFTB2 and spin-purified ΔDFTB2 (mio-1-1) on the identical original 19-state geometries. TD assignments use a pure spatial irrep and the within-irrep ordinal of the fixed DFT reference. ΔDFTB is restricted to the first global reference singlet and requires the final broken-symmetry determinant to have the target spatial irrep. These are symmetry/ordinal comparisons without cross-method transition-overlap qualification. All energies are in eV. "SCC" denotes an incomplete archived calculation; "higher" denotes an inapplicable higher reference state.

| Molecule | State | TBE | MOUSE | TD-DFTB | ΔDFTB |
|---|---|---|---|---|---|
| acetaldehyde | $n\pi^*$ | 4.304 | 4.277 | 4.531 | 4.766 |
| acetone | $n\pi^*$ | 4.468 | 4.338 | 4.629 | 4.922 |
| acrolein | $n\pi^*$ | 3.723 | 3.771 | 3.096 | 3.576 |
| cyclopentadienone | $n\pi^*$ | 2.951 | 2.953 | 2.318 | SCC |
| formamide | $n\pi^*$ | 5.644 | 5.648 | 5.651 | 6.002 |
| glyoxal | $n\pi^*$ | 2.877 | 2.752 | 2.346 | 2.621 |
| maleimide | $n\pi^*$ | 3.810 | 3.643 | 2.729 | 3.134 |
| nitrosomethane | $n\pi^*$ | 1.952 | 1.721 | 2.068 | 2.658 |
| nitroxyl | $n\pi^*$ | 1.743 | 1.485 | 1.729 | 2.336 |
| pyranone | $n\pi^*$ | 4.016 | 4.076 | 3.436 | 4.012 |
| pyrazine | $n\pi^*$ | 4.144 | 3.932 | 3.767 | 4.086 |
| pyrimidine | $n\pi^*$ | 4.451 | 4.218 | 4.277 | 4.488 |
| tetrazine | $n\pi^*$ | 2.462 | 2.060 | 2.387 | 2.713 |
| uracil | $n\pi^*$ | 4.935 | 4.885 | 3.809 | SCC |
| uracil | $\pi\pi^*$ | 5.381 | 5.686 | 4.980 | higher |
| cyanoformaldehyde | $n\pi^*$ | 3.817 | 3.989 | 3.946 | 4.274 |
| cyclopropenone | $n\pi^*$ | 4.274 | 4.482 | 3.342 | 4.147 |
| imidazole | $n\pi^*$ | 6.715 | 6.180 | 9.569 | higher |
| pyridazine | $n\pi^*$ | 3.830 | 3.530 | 3.511 | 3.947 |

## 14 Complete cyclopentadiene multi-CSF Hessian check

All 33 Cartesian columns pass through the public surface, with maximum Hessian error $1.2964 \times 10^{-7}$ $E_\mathrm{h}$/bohr$^2$ and maximum energy–gradient difference $9.0270 \times 10^{-9}$ $E_\mathrm{h}$/bohr. Accepted anchor and all 66 signed energy/gradient evaluations are retained. A separate lightweight reduction of the transferred checkpoints reproduces every reported error. The cold analytic Hessian call takes 250.27 s, with peak process RSS 878220 KiB; this timing begins after model loading and includes direct state relaxation and all three ground-reference derivative terms. The 66 displaced energy/gradient calls take 465.47 s in total, with median 7.05 s (range 4.07–11.27 s); thus the measured finite-difference Hessian cost is 1.86 times the analytic call for this prepared model. Both timings exclude DFT/model preparation and do not constitute a TDA comparison.

Both furan and cyclopentadiene stored Hessians are symmetric and satisfy the translational acoustic identity and nonstationary rotational-covariance identity $H(\Omega R) = \Omega g$. The largest translation residual is below $2 \times 10^{-15}$ $E_\mathrm{h}$/bohr$^2$, the largest rotation residual below $4 \times 10^{-11}$ $E_\mathrm{h}$/bohr, and the net torque below $3 \times 10^{-15}$ $E_\mathrm{h}$. These identities do not imply a molecular minimum or an experimentally accurate vibrational spectrum.

Table S8: Complete 33-column cyclopentadiene total-excited-energy Hessian check through the public multi-CSF surface. The fixed rank-two model includes native S0 restoration and both corrected ground subtractions. Each signed displacement is independently reoptimized from the accepted anchor; central step is $2 \times 10^{-4}$ bohr. Units are $E_h$/bohr and $E_h$/bohr$^2$.

| Cartesian column (zero based) | Energy–gradient error | Max. Hessian error |
|---|---|---|
| 0 | 2.66e-11 | 7.48e-09 |
| 1 | 5.33e-11 | 4.07e-09 |
| 2 | 1.60e-09 | 1.20e-08 |
| 3 | 8.88e-11 | 1.16e-08 |
| 4 | 5.02e-09 | 1.89e-08 |
| 5 | 9.03e-09 | 1.92e-08 |
| 6 | 3.55e-11 | 1.16e-08 |
| 7 | 5.03e-09 | 1.89e-08 |
| 8 | 8.94e-09 | 1.92e-08 |
| 9 | 5.33e-11 | 1.30e-07 |
| 10 | 6.93e-09 | 1.58e-08 |
| 11 | 5.95e-09 | 2.52e-08 |
| 12 | 8.88e-11 | 1.30e-07 |
| 13 | 7.13e-09 | 1.58e-08 |
| 14 | 5.83e-09 | 2.52e-08 |
| 15 | 3.54e-09 | 6.84e-09 |
| 16 | 1.78e-10 | 1.52e-09 |
| 17 | 8.95e-10 | 4.41e-09 |
| 18 | 3.86e-09 | 6.83e-09 |
| 19 | 2.00e-10 | 1.52e-09 |
| 20 | 1.03e-09 | 4.42e-09 |
| 21 | 3.11e-11 | 3.04e-09 |
| 22 | 6.71e-09 | 1.88e-08 |
| 23 | 4.86e-10 | 2.95e-09 |
| 24 | 9.77e-11 | 3.03e-09 |
| 25 | 6.63e-09 | 1.88e-08 |
| 26 | 5.66e-10 | 2.95e-09 |
| 27 | 1.60e-10 | 5.83e-09 |
| 28 | 4.23e-10 | 2.99e-09 |
| 29 | 3.66e-09 | 1.01e-08 |
| 30 | 1.02e-10 | 5.83e-09 |
| 31 | 5.43e-10 | 2.99e-09 |
| 32 | 3.68e-09 | 1.01e-08 |

## 15 Actual DFT-reference rank and subspace test

The same three rank-two molecules are recomputed at uniform internal Cartesian scales 0.98 and 1.02 about the geometric center. The state-identity thresholds and nearest-integer effective-pair rule are unchanged. Two same-symmetry singlet roots must converge at residual tolerance $10^{-7}$ Hartree; independently recomputed full-operator residuals certify the final vectors. The final reference solver explicitly restricts the occupied/virtual pair space to the requested irrep, uses two-vector Davidson increments and linear-dependence threshold $10^{-20}$, and saves positive-root

Ritz checkpoints. Both previously completed neighboring roots can be transported by exact cross-AO overlap. Forbidden-sector numerical components from earlier full-space restart attempts are rejected; their logs and uncertified vectors remain archived and contribute no accepted result. Completed SCF checkpoints are reused. The six final points and baseline reprojection audit ran on Oracle with Python 3.11.15, NumPy 2.3.5 and PySCF 2.14.0; the native extension hash matches the original integration qualification.

The fresh-native-S0 and transported-original-subspace calculations both retain rank two. Baseline compact transition matrices are reproduced within $5 \times 10^{-16}$, before the changed references are analyzed. All six target transition overlaps exceed 0.996 and available-competitor margins exceed 0.972. This is a local reference/identity test with exposed data and limited competing-root coverage, not a full F/Q recalibration, a common-target-geometry energy anchoring test, or a new independent benchmark.

Table S9: Actual displaced-reference rank test. PBE0/aug-cc-pVDZ/grid-3 ground states and two same-symmetry TDA roots are recomputed at uniform Cartesian scales 0.98 and 1.02. A fresh restricted native GFN2 S0 defines the compact projection. $N_{\mathrm{pair}}$ and rank refer to that normalized compact transition. $f$ is coherent cross-geometry transition overlap with the original fixed DFT root; $p$ is the compact projection norm. This does not recalibrate the complete F/Q energy model.

| Molecule | Scale | $N_{\mathrm{pair}}$ | Rank | $f$ | $p$ |
|---|---|---|---|---|---|
| Cyclopentadiene | 0.98 | 1.9927 | 2 | 0.9982 | 0.8256 |
| Cyclopentadiene | 1.02 | 1.9850 | 2 | 0.9985 | 0.8511 |
| Furan | 0.98 | 2.0135 | 2 | 0.9984 | 0.8320 |
| Furan | 1.02 | 2.0071 | 2 | 0.9986 | 0.8585 |
| Phthalimide | 0.98 | 1.5924 | 2 | 0.9962 | 0.9318 |
| Phthalimide | 1.02 | 1.5982 | 2 | 0.9962 | 0.9317 |

## 16 Final frozen cohorts and full-anchor energies

Table S10: Full F/Q and projected-integral recalibration, evaluated at the original common geometry. The original model definition is reproduced exactly at unit scale. All electronic calculations use Oracle with two threads.

| Molecule | Scale | Gap / eV | Change / eV | Residual / $E_h$ |
|---|---|---|---|---|
| Cyclopentadiene | 0.98 | 6.70880 | -0.01134 | 4.5e-15 |
| Cyclopentadiene | 1.02 | 6.72817 | +0.00803 | 5.8e-15 |
| Furan | 0.98 | 6.60398 | -0.06963 | 1.2e-14 |
| Furan | 1.02 | 6.69609 | +0.02248 | 1.3e-14 |
| Phthalimide | 0.98 | 5.18083 | +0.23500 | 6.9e-15 |
| Phthalimide | 1.02 | 5.14244 | +0.19662 | 2.7e-12 |

Table S11: Frozen primary and subsequent challenges, retained separately by identifier I, C and S. S states share a primary-cohort molecule; they are not new molecular holdouts. Dashes denote no qualified energy, never zero error. $^{d}$ flags partial double character in the source; $^{u}$ flags a source reference marked unsafe. Both Criegee states retain the former flag. Errors are against the fixed numerical references.

| ID | Molecule | State | Rank | Ref. | OO | $\delta$OO | $\delta$TDA-Q |
|---|---|---|---|---|---|---|---|
| I001$^{d}$ | Criegee Intermediate | $^1A''$ | 1 | 2.403 | 2.113 | -0.290 | +0.203 |
| I002$^{d}$ | Criegee Intermediate | $^1A'$ | 1 | 3.715 | 1.851 | -1.864 | +2.676 |
| I003 | Cyclobutadiene | $^1B_{1g}$ | 1 | 3.138 | – | – | – |
| I004 | Benzoxadiazole | $^1B_2$ | 1 | 4.606 | 4.491 | -0.115 | – |
| I005 | Hexatriene | $^1B_u$ | 1 | 5.341 | 4.912 | -0.429 | -0.022 |
| I006 | Ammonia | $^1A_1$ | 1 | 6.588 | – | – | – |
| C001 | Furofuran | $^1B_u$ | 1 | 5.489 | 5.523 | +0.034 | +0.492 |
| C002 | Furofuran | $^1A_g$ | 1 | 6.027 | 7.067 | +1.040 | +0.231 |
| S001$^{u}$ | Benzoxadiazole | $^1A_1$ | 2 | 4.911 | 4.587 | -0.324 | +0.567 |
| S002 | Benzoxadiazole | $^1A_2$ | 1 | 5.312 | 4.788 | -0.524 | -0.078 |
| S003 | Benzoxadiazole | $^1B_1$ | 1 | 5.840 | 5.489 | -0.351 | -0.029 |

## 17 Furan physical reference and cyclopentadiene descent

Table S12: Furan internal coordinates versus the same EOM-CCSD stationary-point reference (Table V). Symmetry-equivalent coordinates are averaged.

| Coordinate | EOM-CCSD | MOUSE | Difference | Unit |
|---|---|---|---|---|
| O-Calpha | 1.3780 | 1.3812 | +0.0032 | Å |
| Calpha-Cbeta | 1.4600 | 1.4205 | -0.0395 | Å |
| Cbeta-Cbeta | 1.4640 | 1.4998 | +0.0358 | Å |
| Calpha-H | 1.0850 | 1.0683 | -0.0167 | Å |
| Cbeta-H | 1.0880 | 1.0649 | -0.0231 | Å |
| Calpha-O-Calpha | 109.600 | 106.833 | -2.767 | deg. |
| O-Calpha-Cbeta | 109.600 | 111.935 | +2.335 | deg. |
| Calpha-Cbeta-Cbeta | 105.600 | 104.648 | -0.952 | deg. |
| O-Calpha-H | 116.600 | 117.039 | +0.439 | deg. |
| Calpha-Cbeta-H | 126.500 | 127.743 | +1.243 | deg. |

Table S13: Furan symmetric valence A1 stationary-point in-plane frequencies, in $cm^{-1}$. Reference: EOM-CCSD, Gromov et al., DOI 10.1063/1.1578051, Table VI. Correspondence uses symmetry then descending frequency; cross-method mode-vector overlaps are unavailable. The large B2 discrepancy is retained. The five imaginary out-of-plane model modes have no numerical reference entries in that table.

| Irrep | Order | EOM-CCSD | MOUSE | Difference |
|---|---|---|---|---|
| A1 | 1 | 3332.0 | 3237.0 | -95.0 |
| A1 | 2 | 3303.0 | 3199.6 | -103.4 |
| A1 | 3 | 1446.0 | 1424.3 | -21.7 |
| A1 | 4 | 1407.0 | 1279.9 | -127.1 |
| A1 | 5 | 1079.0 | 1094.6 | +15.6 |
| A1 | 6 | 1066.0 | 1038.6 | -27.4 |
| A1 | 7 | 974.0 | 971.5 | -2.5 |
| A1 | 8 | 769.0 | 736.2 | -32.8 |
| B2 | 1 | 3451.0 | 3230.3 | -220.7 |
| B2 | 2 | 3305.0 | 3200.2 | -104.8 |
| B2 | 3 | 2914.0 | 1440.1 | -1473.9 |
| B2 | 4 | 1404.0 | 1134.8 | -269.2 |
| B2 | 5 | 1261.0 | 1096.8 | -164.2 |
| B2 | 6 | 1053.0 | 991.1 | -61.9 |
| B2 | 7 | 863.0 | 754.4 | -108.6 |

The furan reference is the symmetric A1 valence stationary point, not the nonsymmetric minimum. The independent distorted-point energy–gradient check has error $2.46 \times 10^{-9}$ Hartree/bohr. The cyclopentadiene descendant has 27 positive vibrational frequencies and maximum gradient $2.59 \times 10^{-5}$ Hartree/bohr. A central gradient difference along the softest cyclopentadiene minimum mode, at $2 \times 10^{-4}$ bohr, verifies its analytic Hessian action within $1.33 \times 10^{-10}$ Hartree/bohr$^2$. The retained furan descendant has not reached the same stationarity threshold; no minimum or harmonic spectrum is assigned to it.

# 18 Prepared-property cost, memory, and preparation audit

Table S14: Prepared-property costs on Oracle, two threads. Median wall time over n completed fresh processes (three for small cases, one for the largest three sizes) and maximum whole-process RSS. OO Hessian scope includes energy and total gradient. The three smaller cases use rank one and the three larger cases rank two; size and CSF-rank effects are not separated. Both-parent all-root TDA and a single OO state are different outputs. DFT/FQ preparation and model loading are excluded from the property timer and audited separately.

| Molecule | Property | n | AOs | Rank | Time / s | RSS / MiB |
|---|---|---|---|---|---|---|
| nitroxyl | tda | 3 | 9 | 1 | 0.014 | 152.3 |
| acetaldehyde | tda | 3 | 16 | 1 | 0.057 | 133.8 |
| acetone | tda | 3 | 22 | 1 | 0.155 | 148.2 |
| Furan | tda | 1 | 24 | 2 | 0.224 | 157.2 |
| Cyclopentadiene | tda | 1 | 26 | 2 | 0.303 | 170.2 |
| Phthalimide | tda | 1 | 49 | 2 | 8.731 | 569.3 |
| nitroxyl | energy | 3 | 9 | 1 | 0.286 | 143.5 |
| acetaldehyde | energy | 3 | 16 | 1 | 0.440 | 153.3 |
| acetone | energy | 3 | 22 | 1 | 0.815 | 164.8 |
| Furan | energy | 1 | 24 | 2 | 0.687 | 179.0 |
| Cyclopentadiene | energy | 1 | 26 | 2 | 0.832 | 199.0 |
| Phthalimide | energy | 1 | 49 | 2 | 11.478 | 1017.1 |
| nitroxyl | gradient | 3 | 9 | 1 | 0.301 | 143.6 |
| acetaldehyde | gradient | 3 | 16 | 1 | 0.374 | 152.9 |
| acetone | gradient | 3 | 22 | 1 | 0.500 | 163.4 |
| Furan | gradient | 1 | 24 | 2 | 0.851 | 197.3 |
| Cyclopentadiene | gradient | 1 | 26 | 2 | 1.042 | 210.8 |
| Phthalimide | gradient | 1 | 49 | 2 | 9.292 | 1035.7 |
| nitroxyl | hessian | 3 | 9 | 1 | 0.684 | 145.0 |
| acetaldehyde | hessian | 3 | 16 | 1 | 1.755 | 163.1 |
| acetone | hessian | 3 | 22 | 1 | 3.460 | 194.1 |
| Furan | hessian | 1 | 24 | 2 | 5.104 | 208.9 |
| Cyclopentadiene | hessian | 1 | 26 | 2 | 7.083 | 235.7 |
| Phthalimide | hessian | 1 | 49 | 2 | 42.224 | 1096.7 |

Table S15: Available preparation stages for the fresh molecular cohort, in seconds. DFT includes the ground calculation; reference TDA supplies transition identity and rank. F calibration is a separate recorded stage. The final column is worker wall time after lock and imports through prediction closure, not a sum of stages or necessarily cold preparation. Ammonia reuses an earlier DFT checkpoint. Dashes denote unavailable/not executed stages. Historical six-size benchmark preparation totals are unavailable and are not assigned zero.

| Molecule | AOs | DFT | Ref. TDA | F calib. | Worker |
|---|---|---|---|---|---|
| Criegee Intermediate | 14 | 2.20 | 8.31 | 1.52 | 13.66 |
| Cyclobutadiene | 20 | 4.10 | 10.50 | – | 14.75 |
| Benzoxadiazole | 40 | 38.13 | 73.36 | 77.17 | 249.05 |
| Hexatriene | 32 | 21.37 | 45.07 | 36.35 | 115.60 |
| Ammonia | 7 | 0.51 | 0.88 | – | 0.99 |
| Furofuran | 36 | 25.57 | 101.69 | 50.16 | 253.83 |

## 19 Complete phthalimide analytic Hessian

The complete $48 \times 48$ matrix is obtained with the same Hamiltonian, orbital stationarity and spin-adapted CSF definitions. A first complete profiled evaluation takes 97.08 s. Batched response and reuse of the independent orbital/CI block reproduce that matrix within $1.32 \times 10^{-12}$ Hartree/bohr$^2$. Along the most negative Cartesian Hessian eigenvector, central differences of relaxed gradients at $2 \times 10^{-4}$ and $10^{-4}$ bohr have maximum errors $2.7684 \times 10^{-4}$ and $6.8879 \times 10^{-5}$ Hartree/bohr$^2$, a ratio of 4.019. Richardson extrapolation gives $4.42 \times 10^{-7}$ Hartree/bohr$^2$ error. This step-convergence test uses finite differences only for validation; the production Hessian is analytic. The maximum translational identity residual is $5.59 \times 10^{-15}$ Hartree/bohr$^2$ and the nonstationary rotational-covariance residual is $5.56 \times 10^{-12}$ Hartree/bohr. Earlier censored timings and all displaced-state checkpoints are retained.

Table S16: Analytic derivative acceleration on the same prepared models: old and current energy-plus-gradient and energy-plus-gradient-plus-Hessian wall times in seconds. Small-case values are medians of three fresh processes; large cases have one. Earlier censored phthalimide timing is retained as a lower bound. Energy models, ranks, weights, scaling and derivative definitions are unchanged.

| Molecule | Old E+G | Old E+G+H | New E+G | New E+G+H |
|---|---|---|---|---|
| nitroxyl | 0.319 | 0.971 | 0.301 | 0.684 |
| acetaldehyde | 0.867 | 8.493 | 0.374 | 1.755 |
| acetone | 2.063 | 62.848 | 0.500 | 3.460 |
| Furan | 3.308 | 104.156 | 0.851 | 5.104 |
| Cyclopentadiene | 5.248 | 251.653 | 1.042 | 7.083 |
| Phthalimide | 71.118 | $\geq 600$ | 9.292 | 42.224 |

## 20 Bounded energy-model diagnostics

Increasing the furofuran space to fixed rank two gives 5.39274 and 6.86288 eV for the two challenge states, against fixed references 5.489 and 6.027 eV. The corresponding baseline rank-one values are 5.52261 and 7.06733 eV. Thus a larger singles-CSF space only partly reduces the Ag error. A rank-three continuation is closed at its last accepted homotopy checkpoint without assigning an endpoint energy. Fixed-rank manifests explicitly preserve the requested rank instead of reapplying the automatic NTO rank rule.

A separate bounded diagnostic replaces only the F-parent linear reference by the projected converged-DFT-ground Fock operator, preserving all response terms, weights, scaling and zero-offset policy. Its primary-cohort MAE is 0.58854 eV and its two-state furofuran MAE is 0.47656 eV. This construction was frozen before the primary baseline labels were opened; furofuran labels were already exposed. The modest improvement does not establish a universal correction or justify promoting this diagnostic over the frozen baseline. Prediction locks and exposure records are retained separately.

# 21 Prospective test of a spin-adapted frontier configuration extension

The remaining energy-model error motivated a separate research extension of the orbital-optimized model. This extension is not substituted for the frozen production model in the preceding tables. Its native GFN2 ground spatial orbitals are retained, while the reference one-body levels combine the projected CAM-B3LYP Fock Rayleigh quotients and qualified full-basis orbital energies with the unchanged weights $1/2, 1/2$. The response exchange scale remains $\lambda = 0.9$; orbital gaps are not multiplied by this scale. The same prepared reference and native spatial orbitals define the TDA control. The minimum individually matched native-orbital overlap is greater than $1 - 10^{-7}$ in every control calculation. Thus the comparison does not use full-basis DFT spatial orbitals for TDA.

Optimized, ground-orthogonal ROKS singlets and their paired-double configurations are supplemented by the spin-adapted configurations of excitation rank at most two in the highest-two-occupied/lowest-two-virtual space. An auxiliary CSF with absolute overlap at least 0.7 with a retained CSF is omitted as redundant. The two four-open-shell singlets are exact $S = 0$ Clebsch–Gordan combinations; they are not broken-symmetry determinants relabeled as singlets. Their same-occupation block is obtained from the SU(2)-invariant Heisenberg completion of the quadratic spin functional. Specifically, with singly occupied site densities $D_i$ and feature moments $q_i$, $M_{ij} = q_i^{\mathsf{T}} K_s q_j$, the block is the singlet projection of

$$E_c + \tfrac{1}{2} \sum_i M_{ii} + 4 \sum_{i<j} M_{ij} \mathbf{S}_i \cdot \mathbf{S}_j. \tag{S1}$$

This matches every collinear spin diagonal and the two-spin ROKS limit, without adding a fitted exchange or excitation-energy parameter. Matrix elements between different occupations use singular-safe Slater–Condon algebra and the overlap-weighted mean of the diagonal functional residuals. This is a distinct approximate state-interaction functional, rather than an exact multi-determinant form of CAM-B3LYP.

The reported excitation energy is the selected full state-interaction excited eigenvalue minus the independently stationary ground-functional energy. This is an explicitly state-specific composite reference: the auxiliary lowest state-interaction eigenvalue is *not* silently reset to the ground-functional energy. Its additional lowering is recorded separately and ranges from 0.050 to 2.428 eV in this pilot. Consequently, these results cannot be interpreted as gaps between eigenvalues of one balanced ground-and-excited-state Hamiltonian. No scalar excitation offset is fitted or applied, but this ground-reference approximation remains a material modeling limitation.

The excitation-rank hierarchy was examined on nine already exposed states. Rank two was selected after that development analysis; those nine states are therefore not blind validation. The candidate was then frozen in commit `7666f7a` before its acetone and glyoxal predictions. Both molecules and their labels had appeared in older campaigns, so this is a prospective new-model check on historically exposed molecules, not a truly blind test. Across eleven states the MAE is 0.217 eV and the maximum absolute error is 0.552 eV, compared with 1.163 and 3.660 eV for the identical-kernel native-orbital TDA control. All eleven states meet the fixed wavefunction-assignment gates. However, the two prospective states alone give MAE 0.385 eV, compared with 0.420 eV for TDA. Glyoxal is underestimated by 0.552 eV; hence the frozen 0.30-eV-MAE/0.50-eV-maximum accuracy gate is not passed. The much larger improvement on the exposed development set must not be presented as a universal correction of lone-pair excitations or as an independently validated advantage over all calibrated TDA models.

The analytic nuclear response is defined for a prepared anchored PES. Auxiliary canonical orbitals and physical off-diagonal operators are frozen in the co-rotating orthonormal AO frame; the target ROKS and ground orbitals retain their complete first and second implicit KKT response. Native diagonal energies retain explicit geometry and density response. Slater–Condon matrix

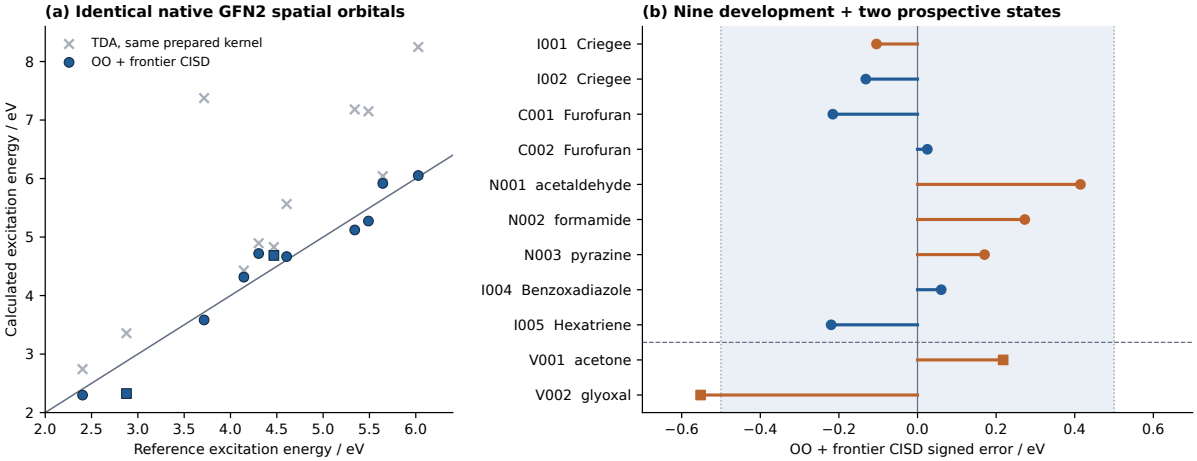



Figure 1: Exploratory spin-adapted frontier CISD extension. (a) Predictions and the same-kernel TDA control use the identical native GFN2 ground spatial orbitals and prepared one-body reference. (b) Signed errors; orange denotes $n \rightarrow \pi^*$ states and blue denotes $\pi \rightarrow \pi^*$ states. Squares below the dashed line identify prospective predictions on historically exposed molecules. The shaded region is the unchanged $\pm 0.5$ eV maximum-error gate, which glyoxal exceeds.

Table S17: Pilot vertical excitation energies in eV. The last two rows were predicted after freezing this candidate.

| ID | Molecule | Reference | OO+CISD | TDA |
|---|---|---|---|---|
| I001 | Criegee Intermediate | 2.403 | 2.299 | 2.743 |
| I002 | Criegee Intermediate | 3.715 | 3.583 | 7.375 |
| C001 | Furofuran | 5.489 | 5.274 | 7.149 |
| C002 | Furofuran | 6.027 | 6.052 | 8.248 |
| N001 | acetaldehyde | 4.304 | 4.719 | 4.894 |
| N002 | formamide | 5.644 | 5.917 | 6.040 |
| N003 | pyrazine | 4.144 | 4.315 | 4.426 |
| I004 | Benzoxadiazole | 4.606 | 4.666 | 5.565 |
| I005 | Hexatriene | 5.341 | 5.121 | 7.182 |
| V001 | acetone | 4.468 | 4.686 | 4.829 |
| V002 | glyoxal | 2.877 | 2.325 | 3.356 |

elements, overlap metrics, the four-spin block, and the generalized eigenvalue are differentiated analytically. Auxiliary orbitals are not re-canonicalized at displaced geometries. This distinction defines the PES being validated, rather than omitting canonical-orbital response from a claimed fully re-canonicalized model. Independent displaced, reoptimized energies are used only to verify the resulting analytic derivatives.

The Criegee target ROKS states can also be recovered from the four canonical frontier promotions through the ground-program frontend, with no TDA amplitude reads and no new DFT or SCC call. The exact CSF overlaps with the retained states differ from unity by less than $10^{-13}$ and the energy differences are below $4 \times 10^{-12}$ eV. The corresponding furofuran recovery has CSF overlaps within $10^{-13}$ of unity and energy differences below $6 \times 10^{-12}$ eV. This verifies a TDA-free initialization route for these four states; it does not establish coverage of CT or Rydberg states outside the chosen compact orbital space.

Complete analytic Cartesian Hessians of this anchored extension have now been assembled for both Criegee states (15 coordinates) and both furofuran states (36 coordinates). Independent reoptimized energies at displacements of $10^{-3}$ and $5 \times 10^{-4}$ bohr in two directions verify the analytic first and second derivatives. Richardson extrapolation of these validation energies gives maximum gradient errors of $1.12 \times 10^{-10}$ and $1.31 \times 10^{-11}$ Hartree/bohr for Criegee and furofuran, respectively, and directional curvature errors of $1.89 \times 10^{-7}$ and $4.03 \times 10^{-7}$ Hartree/bohr$^2$. The analytic Hessian construction itself uses no displaced electronic calculation. Raw Hessian asymmetry is at most $1.16 \times 10^{-13}$ Hartree/bohr$^2$, and the largest translational sum-rule residual is $7.20 \times 10^{-13}$ Hartree/bohr$^2$.

All electronic work for this extension ran on verified Oracle or MAC-STUDIO hosts. Criegee takes 44.69 s on MAC-STUDIO with two BLAS threads. Furofuran preserves its already completed sequential columns and completes the remaining columns with eight independent workers, each using two BLAS threads; that parallel completion takes 126.83 s. These differently scheduled measurements are not a common-hardware, common-parallelism scaling benchmark, and the parallel completion excludes the earlier sequential prefix. They also exclude one-time DFT-reference preparation. No vertical-energy speed advantage over TDA is inferred from them.

## 22 Earlier compact selected-CID generation (superseded)

This section preserves the previous model and its qualification. Its two remaining outliers are resolved by the screened symmetry augmentation reported in the following final-model section. The final development step replaces the auxiliary canonical-singles manifold by a compact selected OO-CID space: the requested optimized, ground-orthogonal ROKS singlets, their paired doubles, canonical frontier double excitations, and the two exact four-spin singlets. Unoptimized auxiliary canonical singles are omitted by excitation rank, using the same rule for every molecule. This model-space choice was selected using thirteen exposed states. It is an empirical design choice, not a proof that auxiliary singles always worsen an orbital-optimized calculation. The common one-body reference averages Fock Rayleigh quotients and qualified DFT orbital levels *before* orbital optimization, with weights $1/2, 1/2$. This differs from averaging two separately optimized parent energies. The exchange-response scale remains $\lambda = 0.9$. There is no orbital-gap scaling or fitted excitation offset.

Let $S_{IJ}$ be the CSF metric, $H^0_{IJ}$ the physical Slater–Condon matrix element, and $d_I = E_I - H^0_{II}/S_{II}$. Away from the exact four-spin block defined above, the effective matrix is

$$H_{IJ} = H^0_{IJ} + \tfrac{1}{2} S_{IJ}(d_I + d_J). \tag{S2}$$

The native density functional supplies $E_I$ including spin polarization. The excited-sector basis is projected orthogonal to the moving stationary ground CSF $|0\rangle$, using $|\widetilde{I}\rangle = |I\rangle - |0\rangle S_{0I}/S_{00}$. We solve the resulting generalized eigenproblem and subtract the independently stationary native ground energy. We differentiate this projector as well as the CSF metric; using an unprojected full-SI root and shifting its energy is not equivalent. The off-diagonal construction remains an approximate state-interaction functional; it does not supply a universally balanced multi-determinant ground-state functional.

The selected rule was frozen in commit `467a9e5` before evaluating nitrosomethane and maleimide. The initial fifteen transitions satisfy the unchanged aggregate MAE gate of 0.3 eV and individual error gate of 0.5 eV, with an MAE of 0.220 eV and maximum absolute error 0.498 eV. The two post-freeze errors are $-0.002$ and $+0.402$ eV, respectively, giving 0.202 eV MAE. These molecules appeared in older campaigns: this is prospective validation of the frozen candidate on historically exposed molecules, not a blind study. Glyoxal is within only 0.002 eV of the individual gate; passing that gate must not be interpreted as a universal correction to all $n\pi^*$ errors. The same-reference, same-native-orbital TDA control gives 0.966 eV MAE over these fifteen states. This control isolates the present kernel and is not a claim to outperform every calibrated TDA or sTDA parameterization.

A further three-molecule cohort was registered in commit `f8a38e3` without changing the candidate: pyrimidine tests another diazine substitution pattern, cyclopentadienone a five-membered conjugated cyclic ketone, and pyranone a six-membered carbonyl-containing oxygen heterocycle. Their $n\pi^*$ predictions are 4.637, 3.412 and 4.512 eV, against unchanged references 4.451, 2.951 and 4.016 eV. All three state assignments pass the fixed overlap gates and all individual errors remain below 0.5 eV. The three-state MAE is 0.381 eV, compared with 0.463 eV for the matched native-orbital TDA control. This subset therefore exceeds the 0.3 eV mean-error criterion; its successful individual assignments and maximum-error checks must not be presented as passing every accuracy gate.

Pooling these cases with the original fifteen gave eighteen states in sixteen molecules, with 0.247 eV MAE and 0.498 eV maximum error. The pooled mean and maximum meet the original numerical limits, but all five predictions made after freezing the candidate give 0.309 eV MAE, slightly exceeding the 0.3 eV mean-error limit. These are historically exposed molecules, not a blind set. The increase from 0.220 to 0.247 eV on extension quantifies the remaining transferability limitation without modifying the model or its offsets. The matched TDA control gives 0.882 eV MAE over eighteen states and 0.442 eV on the five post-freeze predictions. Portable MOUSE

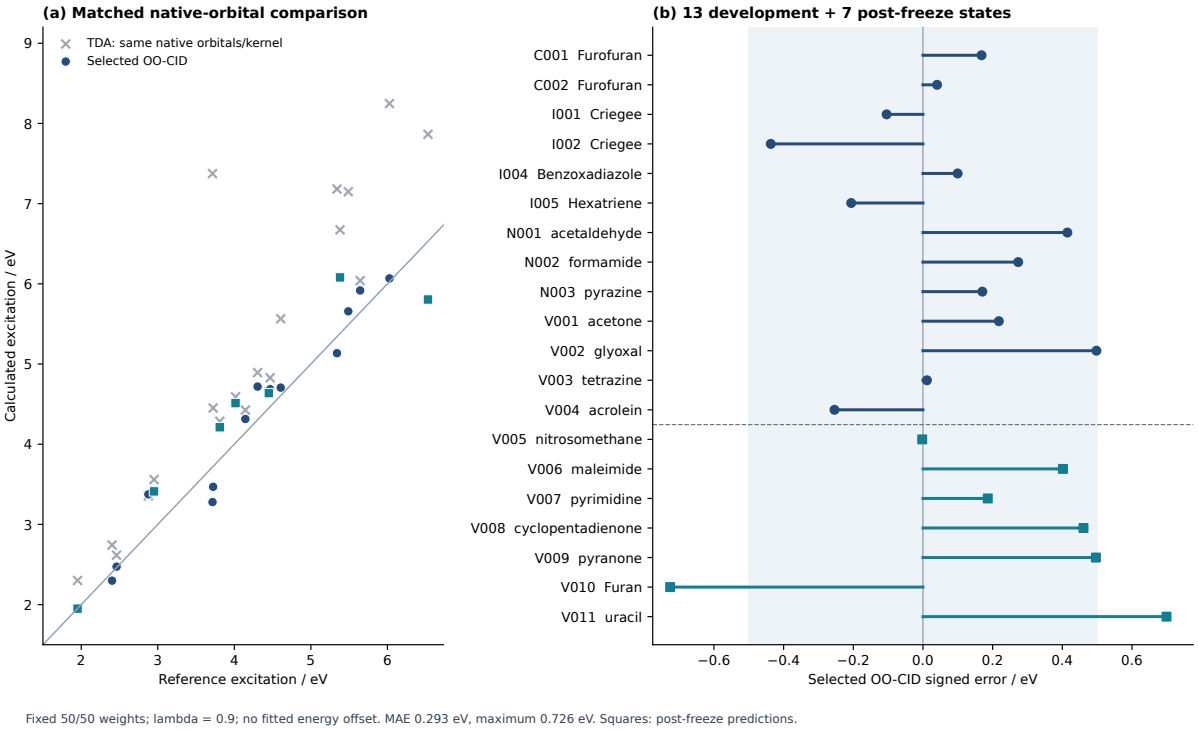



Figure 2: Final selected OO-CID predictions. The shaded error interval is ±0.5 eV. Thirteen states informed model selection; squares identify the seven predictions made after the final candidate freeze. The two last $\pi\pi^*$ errors lie outside the shaded interval. All controls use native GFN2 spatial orbitals.

evaluation reproduces all three new matrix predictions without DFT imports, SCC calls or TDA reads; explicit determinant-sum checks verify $S = 0$ for each new state. No additional nuclear Hessian qualification is claimed for this energy-only extension. Their archived geometries match the existing Jmol structure plates.

Two final $\pi\pi^*$ targets were registered in commit `a474e68`: the $^1A_1$ valence state of furan and the retained uracil target. The frozen model predicts 5.804 and 6.080 eV, against 6.530 and 5.381 eV, respectively. Their errors are −0.726 and +0.699 eV, so both fail the unchanged 0.5 eV individual-error gate. Coherent state-assignment overlaps are 0.842 and 0.994, with margins 0.450 and 0.952; numerical convergence and state assignment pass. Independent portable evaluation reproduces the same energies with no DFT import, SCC call or TDA read, and explicit $S^2$ checks confirm pure singlets. This establishes an energy-model transferability limitation rather than correcting it by numerical convergence. The opposite deviations are separated by 1.425 eV; no common energy offset could bring both inside ±0.5 eV. No offset or state-specific rule is applied.

The complete final set contains twenty states in eighteen molecules: thirteen development states and seven predictions made after freezing the candidate on historically exposed molecules. Its MAE is 0.293 eV and its maximum absolute error is 0.726 eV. The seven post-freeze states give 0.425 eV MAE; the two final $\pi\pi^*$ states alone give 0.712 eV MAE. The pooled mean meets 0.3 eV but the pooled maximum and post-freeze mean do not meet their original limits. The matched native-orbital TDA control gives 0.925 eV overall MAE, 0.691 eV on the seven post-freeze states, and 1.313 eV for these last two targets. Improvement relative to this particular control does not make the final candidate universally accurate. All twenty predictions, including these two accuracy failures, remain in the figure and machine-readable table. No new nuclear-derivative claim is added.

The public class `PreparedNOSIModel` evaluates a portable, checksummed MOUSE surface. LINK selects it with `nosi_50_50`. It uses native ground-program Fock, energy and derivative

machinery, constrained orbital optimization, and a small CSF eigenproblem. It performs no TDA solve, DFT-backend import, or fresh ground-state SCC calculation during evaluation. One-time DFT reference preparation is still required. APOC follows coherent determinant sums, preserving their phases and checking $\langle S^2 \rangle = 0$ using $\|S_+\Psi\|^2$, rather than assigning spin purity to an $M_s = 0$ determinant. Only accepted LINK geometries advance an atomic, model-identified restart; trial geometries leave that restart unchanged.

Complete analytic Hessians of the final selected model are verified for Criegee (two states, 15 coordinates), glyoxal (one state, 18 coordinates), and furofuran (two states, 36 coordinates). Independent reoptimized energy checks at two displacement sizes and directions validate the first and second derivatives; these finite differences are validation only. The largest Richardson-extrapolated errors are $1.33 \times 10^{-10}$ Hartree/bohr for directional gradients and $3.28 \times 10^{-7}$ Hartree/bohr$^2$ for directional curvatures. Both first and second implicit KKT responses, the moving ground projector, metric and generalized eigenvector response are retained. Physical operators and auxiliary orbitals remain fixed in the co-rotating orthonormal AO frame; this precisely defines an anchored PES, not a re-canonicalized model. A fresh-process public-API Hessian reproduces the Criegee research matrices within $9.1 \times 10^{-13}$ Hartree/bohr$^2$, taking 40.82 s with two BLAS threads and 262 MiB peak RSS on MAC-STUDIO. Its energy-only evaluation takes 0.135 s. These timings exclude preparation and do not establish a vertical speed advantage over TDA. The earlier 42.22 s phthalimide Hessian belongs to the preceding production model, not this final selected candidate.

This completes the finite extension and automatic derivative integration, while the final $\pi\pi^*$ accuracy failures remain unresolved. It does not demonstrate an independent broad CT/Rydberg benchmark, final-model excited minima and frequencies against coupled-cluster theory, or universal chemical transferability. Those claims require additional evidence beyond this compact valence-state set.

| ID | Molecule | Reference | OO-CID | TDA |
|---|---|---|---|---|
| C001 | Furofuran | 5.489 | 5.657 | 7.149 |
| C002 | Furofuran | 6.027 | 6.068 | 8.248 |
| I001 | Criegee Intermediate | 2.403 | 2.299 | 2.743 |
| I002 | Criegee Intermediate | 3.715 | 3.278 | 7.375 |
| I004 | Benzoxadiazole | 4.606 | 4.706 | 5.565 |
| I005 | Hexatriene | 5.341 | 5.135 | 7.182 |
| N001 | acetaldehyde | 4.304 | 4.719 | 4.894 |
| N002 | formamide | 5.644 | 5.917 | 6.040 |
| N003 | pyrazine | 4.144 | 4.315 | 4.426 |
| V001 | acetone | 4.468 | 4.686 | 4.829 |
| V002 | glyoxal | 2.877 | 3.375 | 3.356 |
| V003 | tetrazine | 2.462 | 2.473 | 2.616 |
| V004 | acrolein | 3.723 | 3.469 | 4.452 |
| V005 | nitrosomethane | 1.952 | 1.950 | 2.301 |
| V006 | maleimide | 3.810 | 4.212 | 4.286 |
| V007 | pyrimidine | 4.451 | 4.637 | 4.655 |
| V008 | cyclopentadienone | 2.951 | 3.412 | 3.560 |
| V009 | pyranone | 4.016 | 4.512 | 4.591 |
| V010 | Furan | 6.530 | 5.804 | 7.864 |
| V011 | uracil | 5.381 | 6.080 | 6.673 |

Table S18: Earlier compact selected OO-CID and same-native-orbital TDA values in eV; this model is superseded by the screened symmetry augmentation.

## 23 Screened, symmetry-augmented optimized-target CID

The final development candidate addresses the two remaining $\pi\pi^*$ outliers by changing the configuration space and its interaction, while retaining the native GFN2 orbitals, the same prepared F/Q one-body reference, 50/50 weights, $\lambda = 0.9$, and zero excitation-energy offset. It combines optimized ground-orthogonal ROKS target singlets, their paired doubles, and the original two-occupied/two-virtual frontier doubles in both reflection parities. The added configurations are all spin-singlet doubles from the five highest occupied and five lowest virtual canonical native orbitals that are even under reflection in the molecular plane. Both independent singlet recouplings of every four-open-shell occupation are included. Canonical paired doubles with retained-CSF overlap at least 0.7 are omitted by the unchanged duplicate-removal rule. An exact Gaussian-overlap representation certifies the nuclear reflection and canonical MO parities at preparation. The resulting configuration list is frozen during subsequent nuclear displacement; it is not reclassified when an out-of-plane coordinate changes. If reflection or MO parity cannot be certified, no additional symmetry-sector configuration is selected.

This is one common CSF space and one effective matrix for all requested states of a molecule. No $n\pi^*/\pi\pi^*$ label, reference excitation energy, molecular exception, or fitted damping factor enters the selection. For the planar examples, $\pi\pi^*$ targets are reflection-even, whereas $n\pi^*$ targets are reflection-odd. The symmetry augmentation therefore adds the missing even-sector flexibility without introducing the large odd-sector manifold that overcorrelated several controls in the unrestricted five-pair diagnostic. The choice was made after inspecting the exposed twenty-state development set and was frozen in commit `2d59ab8`. It is a model-development choice, not blind validation or a general prescription for nonplanar molecules, CT, or Rydberg excitations.

Environmental screening is introduced through a static finite-basis partition motivated by constrained RPA.[1] Let $V$ be the bare Coulomb matrix in the orthonormal AO-pair chart, $\Delta$ the positive native occupied–virtual gap matrix, and $T$ the symmetrized transition-density columns weighted by their environmental fraction. With occupied and virtual active projectors $P_h, P_v$, these fractions are $1 - (P_h)_{ii}(P_v)_{aa}$; the projectors span the five-pair frontier and retained optimized hole/particle directions. Our prepared static interaction is

$$W = V - VT(\Delta + T^{\mathsf{T}}VT)^{-1}T^{\mathsf{T}}V. \tag{S3}$$

This explicitly specified finite-basis approximation is not a full frequency-dependent cRPA calculation. The same $W$ acts in both reflection sectors. Its one-body term is normal ordered to the unchanged native reference Fock matrix, $h_W = F_{\mathrm{ref}} - J_W[P_g] + K_W[P_g]/2$. Let $H^W_{IJ}$ and $S_{IJ}$ denote its Slater–Condon matrix and CSF metric. The native spin-polarized functional supplies each diagonal energy $E_I$, and the effective matrix is

$$H_{IJ} = H^W_{IJ} + \tfrac{1}{2}S_{IJ}(d_I + d_J), \qquad d_I = E_I - H^W_{II}/S_{II}. \tag{S4}$$

Every two-dimensional four-spin singlet block is restored to the exact SU(2) completion of the same quadratic spin functional. Excited CSFs are projected orthogonal to the moving stationary ground CSF before solving the generalized eigenproblem. Excitation energies subtract its native stationary energy, not an auxiliary correlated-ground eigenvalue. The overlap-weighted residual remains an approximate multistate functional; it is not a proof of a universally balanced correlated ground-state theory.

All seven $\pi\pi^*$ targets now satisfy the unchanged 0.5 eV individual criterion: their MAE is 0.180 eV and maximum absolute error is 0.349 eV. Furan changes from 5.804 to 6.609 eV against 6.530 eV, and uracil from 6.080 to 5.259 eV against 5.381 eV. The thirteen $n\pi^*$ controls remain within 0.5 eV, with 0.250 eV MAE and 0.498 eV maximum error. Overall, the twenty states give 0.225 eV MAE. These are exposed-data development results: the earlier post-freeze designation does not make them prospective tests of this newly selected model. The matched native-orbital TDA control retains its 0.925 eV MAE, using the same F/Q preparation and native response

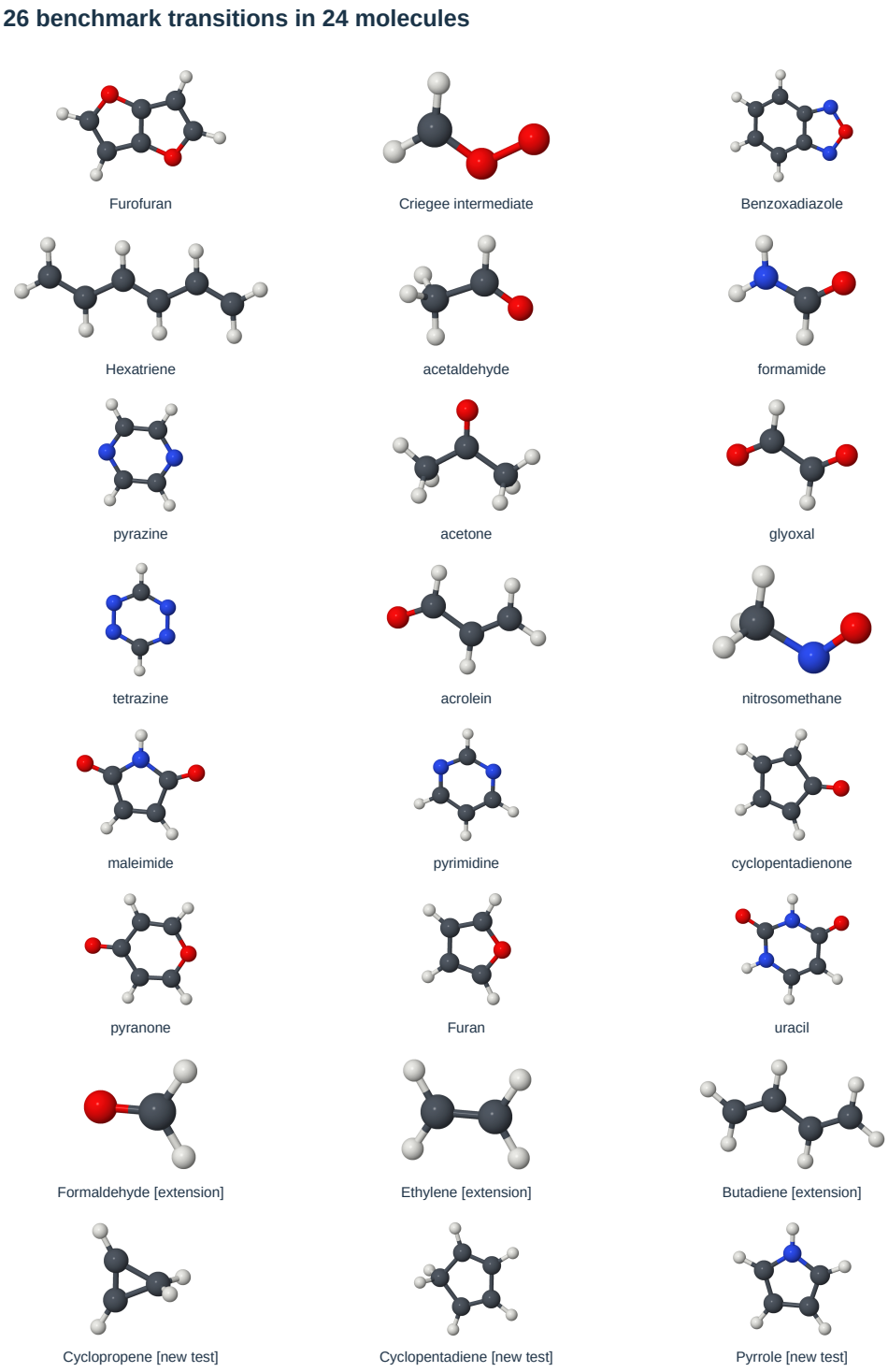



Figure 3: The twenty-four molecular anchors used for the twenty-state development set and its two three-state extensions, rendered with Jmol from the archived three-dimensional coordinates. Only rigid display rotations are applied; panel scales differ. Furofuran contributes two $\pi\pi^*$ states, and Criegee contributes one state of each valence class.

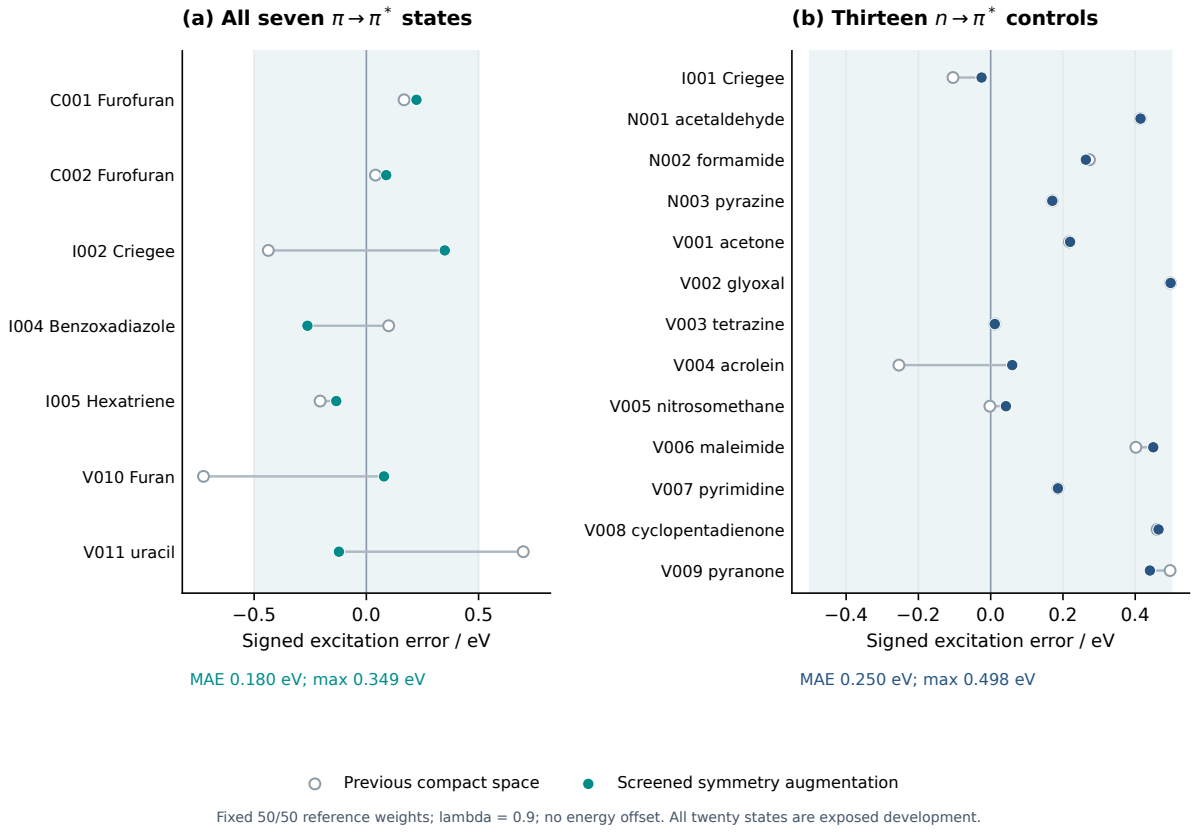



Figure 4: Resolution of the seven $\pi\pi^*$ development targets while retaining the thirteen $n\pi^*$ controls. Open circles show the previous compact-space errors; filled circles show the screened symmetry augmentation. The shaded interval is $\pm 0.5$ eV. All twenty states informed the development assessment; this figure is not a blind-validation result.

kernel; the new screened interaction is an additional ingredient of OO-CID. This comparison does not establish superiority to every calibrated TDA, sTDA-xTB, or TD-DFTB parameterization. All individual values are in the Supporting Information.

The public `PreparedNOSIModel` accepts a checksummed extended model, and LINK retains its `nosi_50_50` selector and accepted-restart contract. Runtime evaluation reuses only ground-engine Fock, energy, spin-polarization, and derivative primitives; it performs no TDA solve, DFT-backend import, or new SCC calculation after preparation. APOC follows the coherent determinant sum and verifies its norm and $\langle S^2 \rangle = \|S_+ \Psi\|^2 = 0$. An individual broken-symmetry $M_s = 0$ determinant is never identified as a spin-pure singlet. The analytic response differentiates the native ground and target KKT systems, all diagonal functionals and spin recouplings, the CSF metric, generalized eigenvectors, and moving ground projector. The prepared $h_W$, $W$, auxiliary orbitals, and configuration list are fixed in the co-rotating orthonormal AO frame. The resulting derivatives are exact for this anchored surface; rebuilding the DFT reference or screening at each geometry defines a different model.

Canonical auxiliary blocks are contracted exactly with frozen-core occupation bitstrings, while optimized-state cross terms retain the singular-safe nonorthogonal Schur/cofactor algebra. Further acceleration caches NumPy's greedy contraction programs, omits only exactly zero derivative terms, and reuses the invariant auxiliary Gram block. Neither an integral nor a derivative term is approximated. Independent canonical-block comparisons agree below $10^{-12}$ Hartree; a molecular mixed-response column agrees with the original implementation within $4.2 \times 10^{-15}$ Hartree/bohr$^2$. The extended public surface and response are committed as `81897cf`; the exact contraction and metric acceleration is `e572d50`. Complete analytic Hessians are

qualified for Criegee (two states, 15 Cartesian coordinates), furan (27 coordinates), and uracil (36 coordinates). Independently reconverged energies at $10^{-3}$ and $5 \times 10^{-4}$ bohr validate both an out-of-plane direction and a mixed internal displacement. Richardson-extrapolated maximum errors are $1.42 \times 10^{-10}$ Hartree/bohr for first derivatives and $3.71 \times 10^{-7}$ Hartree/bohr$^2$ for curvatures. The largest raw Hessian asymmetry is $4.64 \times 10^{-14}$ Hartree/bohr$^2$, and the maximum translational identity residual is $1.43 \times 10^{-13}$ Hartree/bohr$^2$. These are implementation checks of the final anchored surface, not a validation of its equilibrium geometries or physical vibrational frequencies. Fresh-process runtime, spin and restart checks pass for all twenty states in eighteen molecules. Prepared energy-only evaluations take 1.7–9.0 s with two BLAS threads on MAC-STUDIO, excluding model loading and reference preparation. They do not demonstrate a vertical speed advantage over TDA. A direct cold public-API Criegee Hessian reproduces the independently assembled matrices within $2.2 \times 10^{-9}$ Hartree/bohr$^2$. Its complete two-state Hessian evaluation takes 418 s with two BLAS threads and about 1.0 GiB peak RSS; this is the current dense reference implementation, not an end-to-end timing advantage over TDA. The separate column-checkpoint route permits bounded remote parallel validation and reuses completed columns. The final LINK $\pi\pi^*$ test has restart error $3.6 \times 10^{-15}$ Hartree and displaced trial overlap 0.99999996; trial evaluation leaves the accepted checkpoint unchanged.

| ID | Molecule | Reference | Screened OO-CID | TDA |
|---|---|---|---|---|
| C001 | Furofuran | 5.489 | 5.712 | 7.149 |
| C002 | Furofuran | 6.027 | 6.116 | 8.248 |
| I001 | Criegee Intermediate | 2.403 | 2.378 | 2.743 |
| I002 | Criegee Intermediate | 3.715 | 4.064 | 7.375 |
| I004 | Benzoxadiazole | 4.606 | 4.343 | 5.565 |
| I005 | Hexatriene | 5.341 | 5.207 | 7.182 |
| N001 | acetaldehyde | 4.304 | 4.719 | 4.894 |
| N002 | formamide | 5.644 | 5.907 | 6.040 |
| N003 | pyrazine | 4.144 | 4.315 | 4.426 |
| V001 | acetone | 4.468 | 4.688 | 4.829 |
| V002 | glyoxal | 2.877 | 3.375 | 3.356 |
| V003 | tetrazine | 2.462 | 2.473 | 2.616 |
| V004 | acrolein | 3.723 | 3.783 | 4.452 |
| V005 | nitrosomethane | 1.952 | 1.994 | 2.301 |
| V006 | maleimide | 3.810 | 4.260 | 4.286 |
| V007 | pyrimidine | 4.451 | 4.637 | 4.655 |
| V008 | cyclopentadienone | 2.951 | 3.416 | 3.560 |
| V009 | pyranone | 4.016 | 4.457 | 4.591 |
| V010 | Furan | 6.530 | 6.609 | 7.864 |
| V011 | uracil | 5.381 | 5.259 | 6.673 |

Table S19: Final screened symmetry-augmented OO-CID predictions and matched native-orbital TDA values in eV. All twenty targets are exposed development data.

## 24 Structures of all studied molecules

The following Jmol renderings use archived three-dimensional molecular coordinates. Inclusion identifies a studied molecule and does not imply that every requested electronic state is representable or accurate.

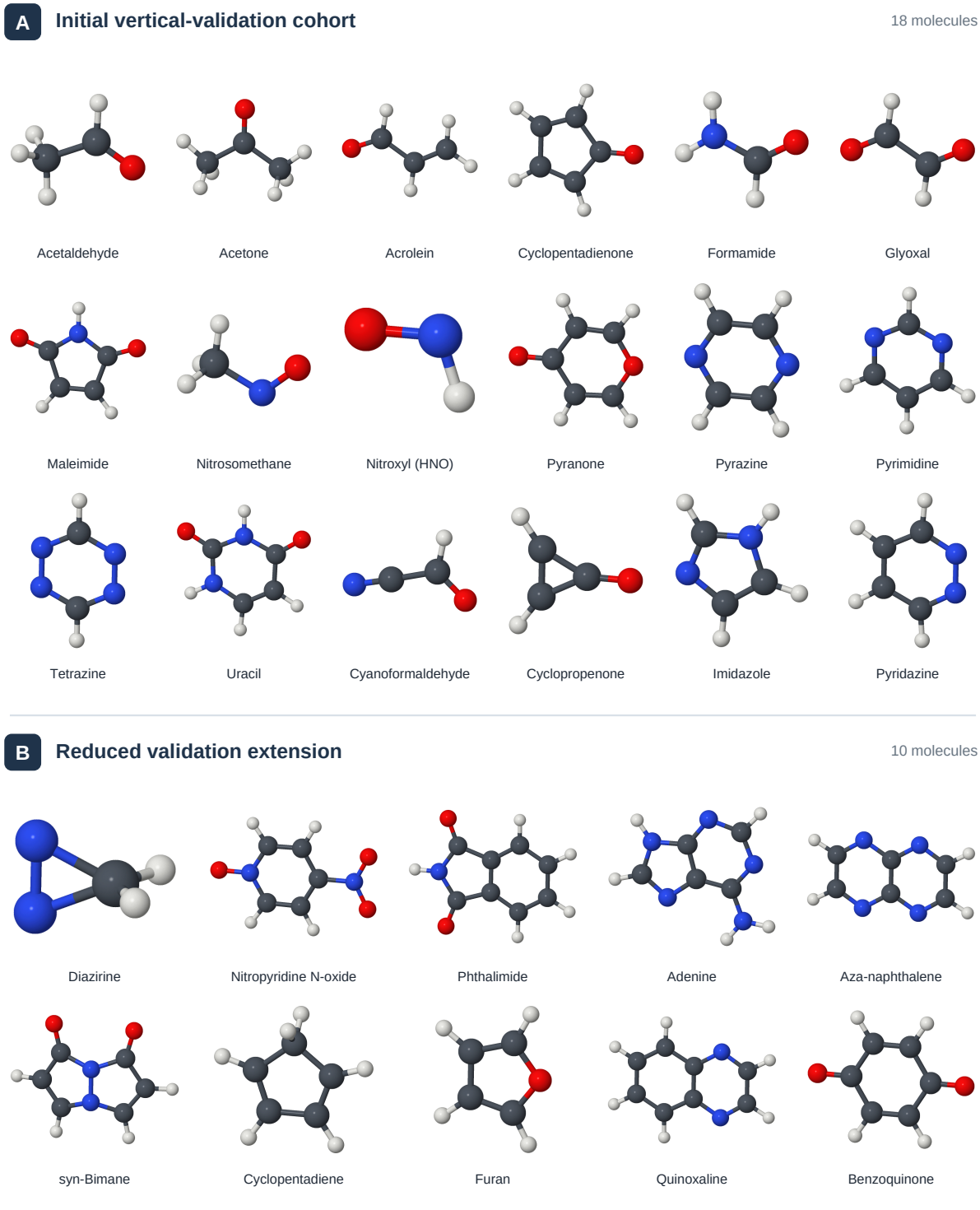



Figure 5: Archived three-dimensional molecular structures, plate 1.

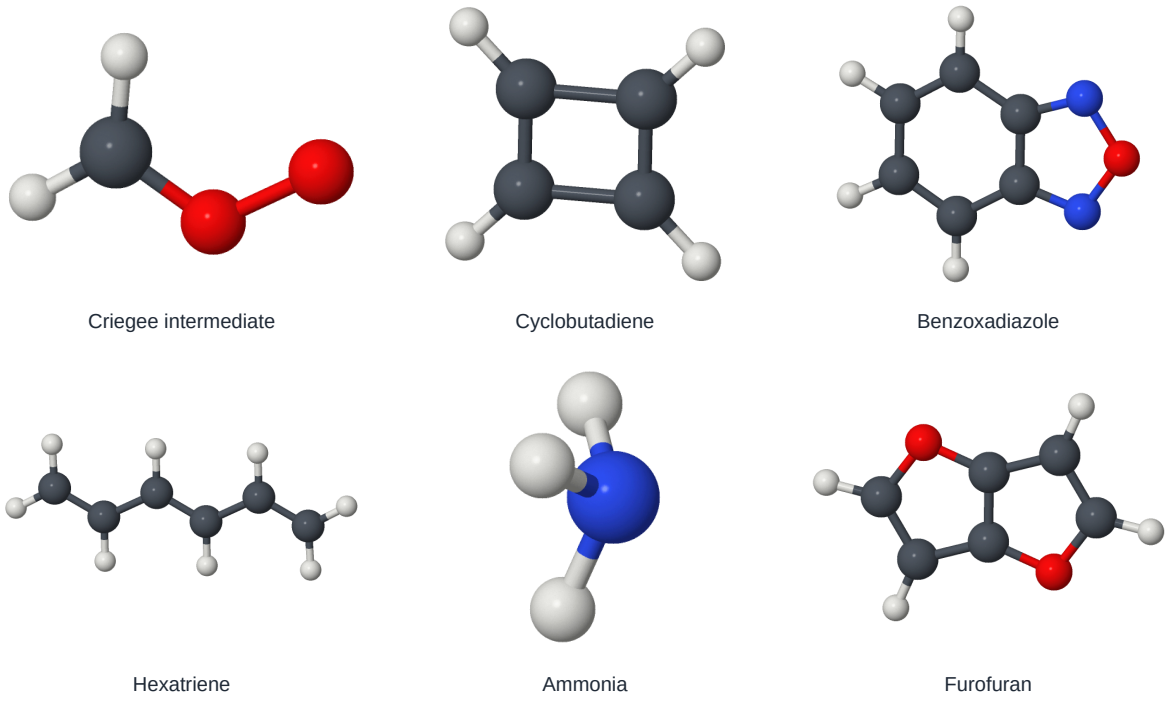



Figure 6: Archived three-dimensional molecular structures, plate 2.

# 25 Closing prospective, anchor and physical validation

This section concerns the final screened-even energy and its unchanged 50/50 weights, exchange scale 0.9 and zero energy offset. It is distinct from the earlier model generations above. All three new fixed predictions were saved before their QUEST scoring labels were opened. The state-seed amendment and the bibliographic exposure of an approximate formaldehyde vertical energy after its prediction are disclosed in `data/closing_validation/preparation_amendment.json`. No new score is used to alter the model.

| Molecule | Reference | OO-CID | TDA | OO error |
|---|---|---|---|---|
| Formaldehyde | 3.966 | 4.285 | 4.650 | +0.319 |
| Ethylene | 7.897 | 9.079 | 11.430 | +1.182 |
| Butadiene | 6.222 | 6.632 | 8.567 | +0.410 |

The prospective MAEs are 0.637 eV for OO-CID and 2.187 eV for the matched native-orbital TDA control. The ethylene error is retained and precludes a universal 0.5 eV claim. All fresh spin-pure runtime and restart checks pass; no DFT backend, TDA input or new SCC driver is required for evaluation.

The eight anchor tests compare excitation gaps at the same displaced geometry. All original target states are retained. The original canonical membership is transported using occupied- and virtual-sector overlap assignments. The maximum absolute gap change is 0.047 eV for Criegee and 0.030 eV for formaldehyde. Re-selecting the configuration space after symmetry breaking is separately recorded: the Criegee space falls from 178 to 13 CSFs and its pi state changes by 0.263–0.273 eV. Integer orbital indices without overlap transport are also retained as a diagnostic, and are not confused with the frozen physical membership. All anchor matrices, reports, mappings and preparation checkpoints remain archived. An initial reporting-only state-index error is corrected by matching `state_id` against the completed eigensystem; no electronic result was refitted.

## 25.1 Final-model minima and physical comparisons

The following internal coordinates belong to the optimized final energy. Formaldehyde references are CC3/aug-cc-pVTZ; acetaldehyde references are CCSDR(3)/def2-TZVPP.[2]

| Molecule / coordinate | Unit | Model | Reference | Difference |
|---|---|---|---|---|
| Formaldehyde / C-O | Angstrom | 1.280 | 1.326 | -0.046 |
| Formaldehyde / C-H | Angstrom | 1.069 | 1.089 | -0.020 |
| Formaldehyde / H-C-H | degree | 117.010 | 118.300 | -1.290 |
| Formaldehyde / pyramidalization | degree | 24.732 | 36.800 | -12.068 |
| acetaldehyde / C-O | Angstrom | 1.274 | 1.331 | -0.057 |
| acetaldehyde / C-C | Angstrom | 1.507 | 1.508 | -0.001 |
| acetaldehyde / C-Hald | Angstrom | 1.081 | 1.091 | -0.010 |
| acetaldehyde / C-C-O | degree | 123.985 | 115.200 | +8.785 |
| acetaldehyde / C-C-Hald | degree | 114.324 | 119.300 | -4.976 |
| acetaldehyde / pyramidalization | degree | 25.384 | 38.700 | -13.316 |

The frequencies below are model harmonic values and experimental fundamentals, in $cm^{-1}$. Their differences include anharmonicity and are not estimates of error against a harmonic coupled-cluster reference. Chemical-coordinate projection determines the mode index. No scale factor is applied.[3, 4]

| Molecule / mode | Index | Harmonic | Fundamental | Difference |
|---|---|---|---|---|
| Formaldehyde / C-O stretch | 4 | 1540.4 | 1173.0 | +367.4 |
| Formaldehyde / symmetric C-H stretch | 6 | 3107.1 | 2847.0 | +260.1 |
| Formaldehyde / asymmetric C-H stretch | 5 | 3059.8 | 2968.0 | +91.8 |
| acetaldehyde / C-O stretch | 11 | 1588.8 | 1119.0 | +469.8 |

Complete minimum coordinates, all harmonic modes, gradient residuals and Hessian certificates are in `data/closing_validation/`. Inversion splittings and torsional level spacings are excluded from the harmonic comparison. The derivative implementation is tested independently of these physical discrepancies.

### 25.2 Exact dense-response acceleration

The optimized dense residual completion gives the same two-state public Criegee energy, gradients and full Hessians bit-for-bit. The archived cold two-thread times are 418.04 and 397.37 s; peak resident memory is 1.06 and 1.19 GB, respectively. This is a single-run 4.9% reduction, not a robust cost-scaling claim. All ten existing spin, canonical-algebra and legacy-contract tests pass. The new numerical work runs exclusively on the verified Mac Studio; existing completed electronic checkpoints and analytic Hessian columns are preserved on resume.

## 26 Common-reference repair and calibrated bath efficiency

The initial twenty-state model, its original prospective extension and its minimum/frequency data above are retained as distinct model generations. The following complete table uses the corrected reference and the single global bath parameter $\eta = 0.60$. Stars mark furan V010 and furofuran C002, the only states used to set $\eta$. All 23 labels had already been exposed; none is claimed as a blind validation. F/Q weights remain 50/50, exchange scale remains 0.9 and the energy offset is zero. Every state is reported.

| Molecule / state | Family | Reference | Corrected OO | Matched TDA | OO error |
|---|---|---|---|---|---|
| Benzoxadiazole / I004 | $\pi\pi^*$ | 4.606 | 4.382 | 5.482 | -0.224 |
| Butadiene / B003 | $\pi\pi^*$ | 6.222 | 6.208 | 8.156 | -0.014 |
| Criegee Intermediate / I002 | $\pi\pi^*$ | 3.715 | 4.021 | 7.139 | +0.306 |
| Ethylene / B002 | $\pi\pi^*$ | 7.897 | 8.206 | 10.639 | +0.309 |
| Furan / V010 * | $\pi\pi^*$ | 6.530 | 6.048 | 7.111 | -0.482 |
| Furofuran / C001 | $\pi\pi^*$ | 5.489 | 5.400 | 6.771 | -0.089 |
| Furofuran / C002 * | $\pi\pi^*$ | 6.027 | 6.508 | 8.162 | +0.481 |
| Hexatriene / I005 | $\pi\pi^*$ | 5.341 | 4.992 | 6.915 | -0.349 |
| uracil / V011 | $\pi\pi^*$ | 5.381 | 5.087 | 6.370 | -0.294 |
| Criegee Intermediate / I001 | $n\pi^*$ | 2.403 | 2.022 | 2.391 | -0.381 |
| Formaldehyde / B001 | $n\pi^*$ | 3.966 | 3.644 | 3.985 | -0.322 |
| acetaldehyde / N001 | $n\pi^*$ | 4.304 | 4.252 | 4.436 | -0.052 |
| acetone / V001 | $n\pi^*$ | 4.468 | 4.392 | 4.541 | -0.076 |
| acrolein / V004 | $n\pi^*$ | 3.723 | 3.412 | 3.989 | -0.311 |
| cyclopentadienone / V008 | $n\pi^*$ | 2.951 | 2.990 | 3.139 | +0.039 |
| formamide / N002 | $n\pi^*$ | 5.644 | 5.366 | 5.493 | -0.278 |
| glyoxal / V002 | $n\pi^*$ | 2.877 | 3.125 | 3.094 | +0.248 |
| maleimide / V006 | $n\pi^*$ | 3.810 | 4.002 | 4.012 | +0.192 |
| nitrosomethane / V005 | $n\pi^*$ | 1.952 | 2.145 | 2.451 | +0.193 |
| pyranone / V009 | $n\pi^*$ | 4.016 | 4.190 | 4.302 | +0.174 |
| pyrazine / N003 | $n\pi^*$ | 4.144 | 4.139 | 4.252 | -0.005 |
| pyrimidine / V007 | $n\pi^*$ | 4.451 | 4.369 | 4.402 | -0.082 |
| tetrazine / V003 | $n\pi^*$ | 2.462 | 2.551 | 2.694 | +0.089 |

The corrected native-reference TDA is recomputed using native GFN2 spatial orbitals, not DFT spatial orbitals. Complete molecule-level checksummed manifests and fresh-process energy/spin/restart certificates are archived in `qualification/closing_validation_20261101/results/bath_eta060`. The corrected ethylene analytic E/G/H certificate and numerical portable bundle are in `qualification/closing_validation_20261101/portable_corrected/Ethylene`. Its gradient and curvature Richardson discrepancies are $1.73 \times 10^{-11}$ Ha/bohr and $6.56 \times 10^{-8}$ Ha/bohr$^2$. The minimization and physical-frequency tables above are not reassigned to this corrected candidate. An opt-in auxiliary-single schema is also available for configuration-space diagnostics; the selected correction in this table retains the original CID configuration space. The fitted screening efficiency requires further prospective testing; it does not establish a universal error correction.

## References


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[3] V. A. Job, V. Sethuraman, and K. K. Innes. The 3500 å transition of formaldehyde-h2, d2, and hd: Vibrational and rotational analyses. *Journal of Molecular Spectroscopy*, 30:365–426, 1969. doi: 10.1016/0022-2852(69)90274-4.

[4] Lynn M. Hubbard, David F. Bocian, and Robert R. Birge. The nature of the $^1n\pi^* \leftarrow S_0$ transition. 4. the first excited singlet state of acetaldehyde. *Journal of the American Chemical Society*, 103:3313–3320, 1981. doi: 10.1021/ja00402a016.

## 27 Spectral-block reference amendment and pigment preparation

The archived directory `data/spectral_blocks/` contains the generic unresolved-orbital rule (`spectral_block_protocol.json`), the original three additional predictions (`additional_scored.json`), and the current complete ledger with matched native-orbital TDA controls (`spectral_block_summary.json`). Pyrrole was exposed before developing this amendment. The pigment protocol specifies primary-source starting geometries, checkpoint reuse, numerical reference preparation, and the remaining state-assignment qualifications. Pigments are not included in these scores.

Table S21: Spectral-block development amendment. All 26 labels are exposed. Native-orbital TDA is recalculated for each corrected model; energies and signed errors are in eV.

| ID | Molecule | Family | Reference | OO-CID | Error | TDA |
|---|---|---|---|---|---|---|
| B001 | Formaldehyde | $n\pi^*$ | 3.966 | 3.636 | -0.330 | 3.985 |
| I001 | Criegee Intermediate | $n\pi^*$ | 2.403 | 1.980 | -0.423 | 2.391 |
| N001 | acetaldehyde | $n\pi^*$ | 4.304 | 4.234 | -0.070 | 4.433 |
| N002 | formamide | $n\pi^*$ | 5.644 | 5.348 | -0.296 | 5.492 |
| N003 | pyrazine | $n\pi^*$ | 4.144 | 4.137 | -0.007 | 4.251 |
| V001 | acetone | $n\pi^*$ | 4.468 | 4.371 | -0.097 | 4.535 |
| V002 | glyoxal | $n\pi^*$ | 2.877 | 3.122 | +0.245 | 3.094 |
| V003 | tetrazine | $n\pi^*$ | 2.462 | 2.549 | +0.087 | 2.693 |
| V004 | acrolein | $n\pi^*$ | 3.723 | 3.369 | -0.354 | 3.938 |
| V005 | nitrosomethane | $n\pi^*$ | 1.952 | 2.140 | +0.188 | 2.449 |
| V006 | maleimide | $n\pi^*$ | 3.810 | 3.994 | +0.184 | 4.004 |
| V007 | pyrimidine | $n\pi^*$ | 4.451 | 4.355 | -0.096 | 4.401 |
| V008 | cyclopentadienone | $n\pi^*$ | 2.951 | 2.971 | +0.020 | 3.121 |
| V009 | pyranone | $n\pi^*$ | 4.016 | 4.181 | +0.165 | 4.302 |
| B002 | Ethylene | $\pi\pi^*$ | 7.897 | 8.205 | +0.308 | 10.569 |
| B003 | Butadiene | $\pi\pi^*$ | 6.222 | 6.204 | -0.018 | 7.987 |
| C001 | Furofuran | $\pi\pi^*$ | 5.489 | 5.363 | -0.126 | 6.675 |
| C002 | Furofuran | $\pi\pi^*$ | 6.027 | 5.610 | -0.417 | 7.598 |
| I002 | Criegee Intermediate | $\pi\pi^*$ | 3.715 | 3.995 | +0.280 | 6.786 |
| I004 | Benzoxadiazole | $\pi\pi^*$ | 4.606 | 4.390 | -0.216 | 5.451 |
| I005 | Hexatriene | $\pi\pi^*$ | 5.341 | 4.955 | -0.386 | 6.803 |
| P001 | Cyclopropene | $\pi\pi^*$ | 6.729 | 6.741 | +0.012 | 7.586 |
| P002 | Cyclopentadiene | $\pi\pi^*$ | 5.540 | 5.492 | -0.048 | 6.799 |
| P003 | Pyrrole | $\pi\pi^*$ | 6.247 | 6.302 | +0.055 | 8.206 |
| V010 | Furan | $\pi\pi^*$ | 6.530 | 6.084 | -0.446 | 7.097 |
| V011 | uracil | $\pi\pi^*$ | 5.381 | 5.103 | -0.278 | 6.339 |

## 28 Large-system orbital-solver qualification

The verified Mac Studio executes every new numerical calculation. The matrix-free orbital Jacobian action agrees with the dense expression within $8.9 \times 10^{-16}$; the determinant constraint action agrees within $1.2 \times 10^{-16}$. After finite orbital perturbations, including an orbital trust radius of 0.001 rad that exercises the Newton/descent subspace, both GMRES and MINRES recover the prepared ethylene and pyrrole roots within $1.1 \times 10^{-14}$ Ha. The real-d harmonic transport agrees with direct polynomial rotation within $5.4 \times 10^{-15}$. Independent directional checks of the first and second frame derivatives give discrepancies $6.6 \times 10^{-12}$ and $9.3 \times 10^{-8}$, respectively.

The full frozen-quadrature correction is checked against the established ethylene functional for arbitrary perturbed spin densities. Energy, Fock, charge response and spin response agree within $2.5 \times 10^{-15}$ in their corresponding atomic units; the completed singlet CI matrix agrees within $3.6 \times 10^{-15}$ Ha. The complete analytic ethylene Hessian, under an invertible complete

response-chart change, agrees within $1.1 \times 10^{-15}$ Ha/bohr$^2$. Fifteen existing algebra, spin and restart-contract tests, two target-subspace invariance/fragmentation tests and three independent observable-phase tests pass. Cached Slater contractions reproduce their direct contractions exactly in the numerical control. Detailed certificates and the qualified implementation snapshot are archived under `qualification/closing_validation_20261101/results/pigments/`.

The carotene 15-CSF matrices and eigensystem are saved separately from the expensive reference arrays. The reported 63.73 s final resumed process reuses its completed CI matrix and must not be interpreted as a total calculation time. PBE0 and CAM reference preparations alone take 1734.10 and 1534.74 s. Full tensor storage and one-time reference preparation remain substantial costs. The current pigment quadrature adapter rejects nuclear-derivative requests; these small-molecule Hessian controls do not constitute pigment gradient or Hessian validation.

## 29 Carotene configuration-space control

Table S22: Carotene vertical configuration-expansion control. Energies in eV; dimensionless oscillator strengths.

| $N$ | $W$ | CSFs | $E_1$ | $f_1$ | $E_2$ | $f_2$ | $E_3$ | $f_3$ |
|---|---|---|---|---|---|---|---|---|
| 2 | 5 | 15 | 3.269 | 6.8319 | 3.680 | 0.005 | 4.558 | 0.011 |
| 4 | 5 | 153 | 2.839 | 2.1153 | 2.860 | 3.660 | 3.407 | 0.077 |
| 5 | 5 | 351 | 2.561 | 0.0112 | 2.728 | 5.448 | 3.032 | 0.065 |
| 5 | 6 | 351 | 2.550 | 0.0100 | 2.728 | 5.449 | 3.021 | 0.061 |
| 6 | 6 | 703 | 2.446 | 0.0092 | 2.671 | 5.308 | 2.879 | 0.064 |
| 6 | 8 | 703 | 2.426 | 0.0076 | 2.673 | 5.294 | 2.857 | 0.075 |
| 7 | 8 | 1275 | 2.373 | 0.0043 | 2.637 | 5.095 | 2.784 | 0.211 |
| 8 | 8 | 2145 | 2.197 | 0.0024 | 2.555 | 4.465 | 2.611 | 0.651 |
| 8 | 10 | 2145 | 2.189 | 0.0023 | 2.559 | 4.109 | 2.605 | 1.016 |
| 9 | 10 | 3403 | 2.145 | 0.0019 | 2.526 | 2.553 | 2.573 | 2.533 |
| 10 | 10 | 5151 | 2.084 | 0.0016 | 2.467 | 0.799 | 2.532 | 4.229 |
| 11 | 11 | 7503 | 1.956 | 0.0012 | 2.349 | 0.137 | 2.484 | 4.787 |
| 10 | 12 | 5151 | 2.070 | 0.0014 | 2.453 | 0.491 | 2.535 | 4.536 |
| 11 | 12 | 7503 | 1.954 | 0.0011 | 2.345 | 0.134 | 2.483 | 4.786 |
| 12 | 12 | 10585 | 1.953 | 0.0011 | 2.344 | 0.136 | 2.481 | 4.790 |
| 13 | 13 | 14535 | 1.950 | 0.0012 | 2.338 | 0.132 | 2.478 | 4.796 |
| 10 | 14 | 5151 | 2.052 | 0.0013 | 2.432 | 0.330 | 2.530 | 4.683 |
| 10 | 16 | 5151 | 2.028 | 0.0012 | 2.405 | 0.220 | 2.520 | 4.772 |
| 11 | 16 | 7503 | 1.911 | 0.0010 | 2.293 | 0.076 | 2.470 | 4.808 |
| 12 | 16 | 10585 | 1.910 | 0.0010 | 2.292 | 0.077 | 2.468 | 4.813 |
| 13 | 16 | 14535 | 1.909 | 0.0010 | 2.291 | 0.078 | 2.466 | 4.817 |

The brightest state is not constrained to retain the same energy index. At $N = 9$, the frontier target is shared by roots 2 and 3, each below the original 0.49 individual-weight threshold. Their joint coverage is 0.805; retaining the target subspace resolves assignment without changing energies. At $N = 10$, the dominant intensity belongs to root 3. Thus the robust observation is weak $S_1$ below an absorbing pair, rather than a fixed single-root bright label. The final $N = 11, 12, 13$ sequence uses matched $W = N$ and reaches maximum consecutive increments of 0.0054 eV. At fixed $W = 16$, the corresponding maximum is 0.0021 eV. Both satisfy the two-increment 0.02 eV criterion for the prepared rank-two model.

All rows use the same source geometry, F/Q weights 50/50, exchange scale 0.9, bath efficiency 0.60 and zero energy offset. $W$ is the occupied and virtual screening-exclusion window; $N$ is the occupied and virtual CI window. The 15-CSF baseline uses $N = 2$ and $W = 5$. No state-specific parameter is introduced. Spectra were exposed before the expansion. The $W = 6$, $N = 6$ CI contains the $N = 5$ principal block with maximum Hamiltonian discrepancy $3.13 \times 10^{-13}$ Ha and

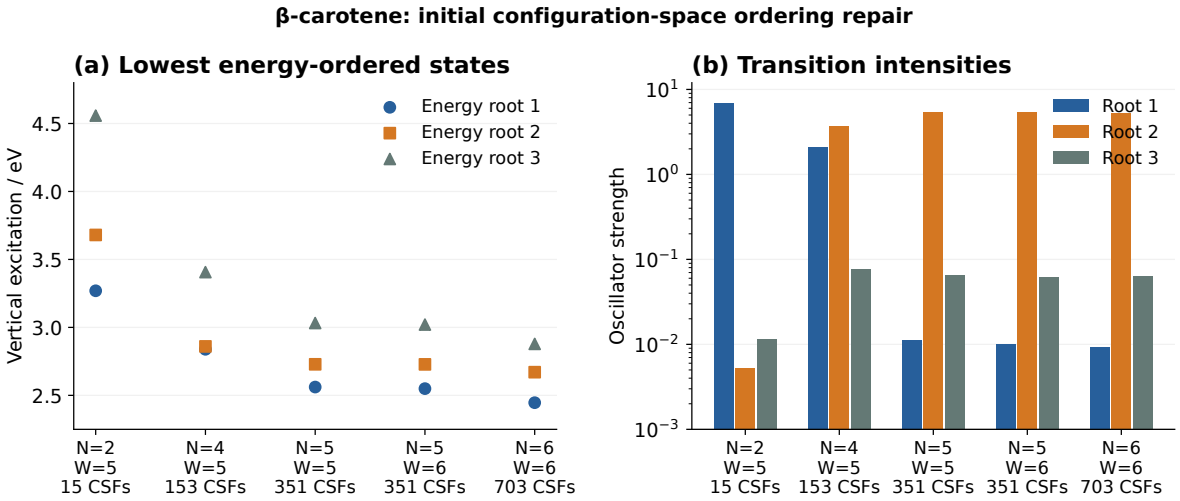



Figure 7: Initial ordering repair from 15 to 703 spin-singlet configurations. Energy-ordered roots are shown with their oscillator strengths; screening window $W$ and CI window $N$ are specified separately. The larger $W = N$ sequence and the fixed-$W = 16$ control are shown in the main text.

overlap discrepancy $2.61 \times 10^{-15}$. The largest $N = 9$ to 10 comparison at $W = 10$ reproduces the 3,403-CSF parent block within $2.28 \times 10^{-13}$ Ha and $2.60 \times 10^{-15}$ in its metric. The absorbing pair has squared principal cosines 0.998 and 0.993. Ground energies are identical and the lowest roots satisfy the Rayleigh–Ritz bound. Changing $W$ changes the screened interaction and is not a same-Hamiltonian variational test. In particular, $W$ specifies which native response channels are excluded, rather than truncating a numerical polarization expansion. At fixed $N = 10$, changing $W = 14$ to 16 shifts the first two gaps by $-0.0233$ and $-0.0269$ eV. The production sequence therefore ties $W$ to the represented CI frontier; robustness to arbitrary mismatched partitions is not claimed. The final fixed-$W = 16$, $N = 12$ to 13 comparison reproduces its 10,585-CSF parent block within $7.7 \times 10^{-13}$ Ha and $6.8 \times 10^{-15}$ in the metric. The absorbing-subspace squared principal cosines are 0.99991 and 0.99986.

Fixed-bra contractions are also compared with direct elements for physical and general fourfold-Hermitian tensors, including up to sixteen active electrons per spin in the physical-tensor cases. Exact sparse spin recoupling is checked on 1,260 off-diagonal elements of the archived 2,145-CSF carotene block, with maximum Hamiltonian difference $1.04 \times 10^{-14}$ Ha and overlap difference $3.71 \times 10^{-13}$. Product cofactors are compared with explicit minors at full and deficient ranks. The exact full-quadrature Hessian refinement is exercised by a small-molecule solve and density-span energy shortcuts are checked against full evaluation. Twenty regressions include the small-molecule analytic Hessian and independent CSF phase checks. Observable contractions share the Hamiltonian determinant ordering and are checked against direct overlaps with its stored metric. Legacy observable arrays are retained separately; energies and CI vectors were reused unchanged when refreshing intensities. Screened reassembly from cached native functional values recovers an archived matrix within $2.0 \times 10^{-11}$ Ha. Three independent checks at each screening window give dielectric residuals below $3.4 \times 10^{-15}$ and auxiliary-factor row errors below $3.3 \times 10^{-18}$. These algebraic controls and the frontier convergence tests address different errors; neither is an experimental accuracy validation.

## 30 Chlorophyllide orbital stationarity and root identity

The prepared chlorophyllide a model is unchanged: F/Q weights 50/50, exchange scale $\lambda = 0.9$, bath efficiency $\eta = 0.60$, and zero energy offset. The initial residual-norm search stagnated away from a root. The successful initialization minimizes the original ROKS energy with an augmented-Lagrangian ground-overlap constraint; the penalty is used only by the optimizer and is absent from reported energies. For the freeze-and-release cases the two promoted open orbitals remain fixed during the first relaxation. Analytic exponential-map pullbacks supply the L-BFGS gradient. Releasing all nonredundant rotations is followed by the full analytic KKT Newton search. The trust basis is projected onto the ground-constraint tangent, and a thin SVD avoids squaring the response condition number.

A converged candidate from the third native promotion initially duplicated the fourth root. The coherent singlet overlap detects this identity ($|S| = 1$). A new initial hole is its orthogonal complement in the same two-dimensional native occupied frontier block. The final search imposes no such excited-state orthogonality constraint. The four retained singlets are independent: their overlap eigenvalues are 0.688173, 0.836139, 1.066701 and 1.408987. All ground overlaps satisfy the original tolerance; Table S23 reports the complete residuals. The already converged fourth target is reused unchanged.

Table S23: Certified chlorophyllide a ROKS stationary roots on the unchanged prepared surface. Target labels identify the frontier initialization; they are not an energy ordering or a claim of pure single-excitation character. Energies precede screened CID and are not final spectral predictions.

| Target | Initial label | $\Delta E$/eV | $\|R\|_\infty$ | $\|o\|$ | Frontier fidelity |
|---|---|---|---|---|---|
| 1 | H→L | 2.849752 | $1.71 \times 10^{-14}$ | $9.86 \times 10^{-15}$ | 0.978234 |
| 2 | H−1 →L | 2.715719 | $4.52 \times 10^{-14}$ | $1.10 \times 10^{-14}$ | 0.982488 |
| 3 | H→L+1 | 3.341362 | $3.95 \times 10^{-12}$ | $2.19 \times 10^{-16}$ | 0.987979 |
| 4 | H−1 →L+1 | 3.648886 | $3.49 \times 10^{-11}$ | $5.92 \times 10^{-13}$ | 0.989335 |

The orbital-stationarity implementation passes 25 algebra, solver, spin, phase and restart tests, twelve small-molecule root-recovery controls and the complete ethylene analytic-Hessian regression. An independent derivative audit at a stalled chlorophyllide checkpoint compares the full analytic moving-frame Jacobian with centered differences for validation only; relative errors are below $4.2 \times 10^{-8}$ and decrease with the step. No finite difference enters the production optimizer. The final four stored roots are independently rechecked without orbital iterations, retraction or multiplier adjustment. No new SCC, TDA or DFT calculation is used by their prepared-model evaluation. Full certificates, checkpoint identities and the duplicate-root rejection are archived in `qualification/closing_validation_20261101/results/chlorophyllide_stagnation_20261102/`. The pigment adapter remains vertical and orbital-response only.

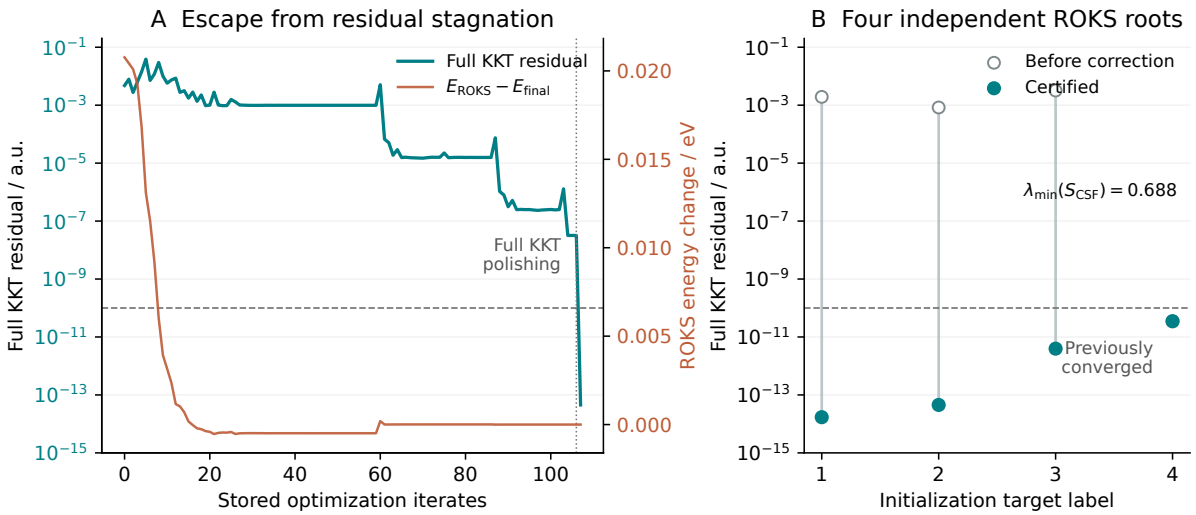



Figure 8: Removal of chlorophyllide ROKS stagnation on the unchanged energy surface. (A) Direct-energy initialization of target 2 and final full KKT polishing. The residual includes ground overlap and need not decrease monotonically during initialization. The orange curve is the original ROKS energy relative to its final value, excluding the numerical constraint penalty. (B) The four initialization targets before correction and after certification. The dashed line is the $10^{-10}$ residual tolerance; the fourth target was already converged. These are orbital-stationarity checks, not nuclear-Hessian or correlated-spectrum qualifications.

## 31 Static self-energy matching: current candidate qualification

The opt-in manifest field `static_self_energy_subtraction` selects the static-matched candidate described in the main text. The primary space contains the optimized singlets and their paired partners; the complete canonical auxiliary complement is retained. The counterterm uses the inverse of the full interacting complementary block, whose positivity and inverse residual are checked. No excitation-energy offset or additional fitted coefficient is introduced. The 50/50 weights, $\lambda = 0.9$ and $\eta = 0.60$ are unchanged.

All 24 public models reproduce the independently assembled energies for 26 states and pass coherent-singlet spin and accepted-restart checks. Their aggregate errors are reported in Table S24. All labels had been exposed before developing this amendment. These results do not constitute blind validation.

Table S24: Static-matching development candidate on the same corrected native orbitals, geometries and 26 exposed reference labels. No parameter is refitted. All errors are in eV.

| Family | States | TDA MAE | Before MAE | Matched MAE | Matched max. |
|---|---|---|---|---|---|
| All | 26 | 0.789 | 0.198 | 0.183 | 0.478 |
| $\pi\pi^*$ | 12 | 1.514 | 0.216 | 0.209 | 0.478 |
| $n\pi^*$ | 14 | 0.168 | 0.183 | 0.160 | 0.411 |

The complete analytic nuclear response differentiates the inverse, ground projector, metric orthogonalization and ground-state energy. Ethylene and furan provide independent controls of this response. Furan has a nonzero static correction of approximately 0.755 eV, so its check also exercises the new counterterm derivatives. Centered scalar-energy Richardson differences are used only to validate the analytic arrays.

Table S25: Current static-matched Hessian qualification on MAC-STUDIO. Errors are maximum independent directional gradient and curvature errors. Times include the prepared public-API call, excluding reference preparation.

| Molecule | Gradient / Ha bohr$^{-1}$ | Curvature / Ha bohr$^{-2}$ | Serial / s | Parallel / s |
|---|---|---|---|---|
| Ethylene | $1.86 \times 10^{-11}$ | $3.07 \times 10^{-8}$ | 9.82 | 6.10 (4 workers) |
| Furan | $1.18 \times 10^{-10}$ | $6.50 \times 10^{-8}$ | 1386.40 | 297.18 (8 workers) |

For both molecules the parallel and serial energy, gradient and complete Hessian arrays agree exactly. Worker processes reuse the prepared native state, converged excited orbitals and multipliers. No worker launches a new SCC, TDA or DFT calculation. LINK uses its existing processor budget for this optional analytic column parallelism. Cold start, accepted restart and displaced-trial rollback are checked with static matching enabled. The earlier-model Hessian certificates elsewhere in this SI remain evidence for those model generations, not additional current-model qualifications.

For chlorophyllide the lowest two static-matched gaps meet the 0.02 eV two-increment criterion over matched frontiers 14, 15 and 16, with a separate fixed-screening control. At frontier 16, retaining twenty roots recovers the native target subspace with minimum coverage 0.527, above the predeclared 0.49 threshold. Coverage is fragmented over ten roots. This algebraic target-coverage check does not establish frontier convergence of the higher spectrum. Pigment nuclear derivatives and independent experimental accuracy are not qualified by these tests.

## 32 Deferred full-DFT companion implementation

The full-integral PySCF DeltaSCF/ROKS implementation and its analytic nuclear-response qualification are intentionally deferred to a separate companion manuscript. They are not part of the semiempirical model, accuracy ledger, timing analysis, or claims made here. The source and qualification archive are retained in the repository for reproducibility and future comparison with ORCA and native PySCF implementations.